Master's Thesis

Flexible Prior Distributions for Stochastic Volatility in Bayesian Structural Vector Autoregressions

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Version note

This version contains minor typographical, grammatical, and formatting corrections to the version submitted and defended at TU Dortmund University. No substantive changes have been made to the content, results, or conclusions.

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1 Introduction

This thesis explores the idea that standard Stochastic-Volatility-identified Bayesian Structural Vector Autoregressions can be expanded with more flexible priors, namely for the variance parameter of the log-volatility, to better model periods of strong, sudden changes in the model variables' behavior.

Structural Vector Autoregressions (VARs) are a common modeling framework in macroeconomics. They allow researchers to decompose the behavior of a system of variables into series of structural shocks, with a variety of identification strategies available to choose from. Stochastic-Volatility in a VAR model describes the idea that the variance of the structural shocks is not constant in time, but follows some random process. For several years now there has been increased recognition of the value of flexibility in VAR analysis, reflecting the intuition that the processes which govern the behavior of these variables are not constant over time. This makes Stochastic Volatility an increasingly popular choice to model the variance of structural shocks: usually the logarithm of the shock variances (the log-volatilities) are specified as following standard independent AR(1) processes. This model is often used to complement Time Varying Parameter VARs (TVP-VARs).

Using SV as an identification strategy has the further advantage of pinning down n structural shocks in a VAR with n variables, while other identification methods are often only able to identify a few choice shocks. However, unlike with other strategies, the identified shocks are not labeled, so an economic interpretation is not automatically attributed to each of them. Yet, SV can be complemented with other identification strategies (Carriero et al., 2024), like sign restrictions (Uhlig, 2005), instrumental variables (Mertens and Ravn, 2013) or narrative restrictions (Antolín-Díaz and Rubio-Ramírez, 2018).

In Bayesian estimation of such SV-VARs, the priors on the variance of log-volatility processes tend to be tight Inverse Gamma priors which assume this variance to be the same for all time points. The question can be raised, however, of whether the usual SV parameterization provides enough flexibility for some situations. Particularly, in a scenario like the one seen during the COVID-19 pandemic. The economic lockdowns in the 2020-2021 period caused unprecedented, sudden, very large shocks that were registered by nearly every macroeconomic indicator. This creates a challenge in

modeling such variables: models that are not flexible enough cannot accommodate this disturbance well and either do not explain or have their results entirely driven by the extreme observations from this period.

Bobeica and Hartwig (2023) look into the challenges the COVID shock brings to inflation modeling with Bayesian VARs. They find, using a 7-variable VAR, that both impulse response functions and forecasts are significantly affected by the inclusion of the pandemic observations. They propose the use of a fat-tailed prior to make their estimates less sensitive to outliers.

TVP-VAR parameters, like in Primiceri (2005), with a time-varying mechanism of immediate transmission of structural shocks to model variables (i.e., a time-varying impact matrix) can be a solution to accommodate the temporary, sharp change in the VAR behavior. But if one believes the actual change is in the shocks themselves, and the mechanism of transmission remains unchanged over time, then the resulting estimated structural shock series would be incorrect, invalidating analyses which look into the historical behavior of shocks (Historical Decompositions, for example).

In the case of time-invariant VAR parameters, the AR(1) process for log-volatility without changes in the parameters over time is clearly not enough, as it does not allow for sudden, very large shocks right after a period of smaller ones.

A way to model these is to allow the log-volatility to have a momentary spike at the time of the disturbance. This can be done, for example through a Markov-switching model for the variance of shocks (Lütkepohl and Woniak (2020)). This allows the model to switch as needed between high and low variance regimes. However, such an approach requires sufficiently many observations in each variance regime to be reliable (Schlaak et al. (2023)), so it is not applicable if the period of large shocks is relatively short, as in the case of the pandemic.

A more flexible way is to let the change be defined through the variance of the log-volatility processes, allowing the variance to be much higher at the point of disturbance. The changing variance can be modeled by explicitly setting a different prior for the parameter at the time of the shock, but this would require deciding on the exact moment and duration of the increased variance period.

The most flexible way is, then, to allow the variance of log-volatility to freely vary with time. Prüser (2021) does just this, although employing a Random Walk process log-volatility instead of AR(1), and integrated into a TVP-VAR. The paper proposes placing a Horseshoe prior on the variance of log-volatility, which provides flexibility while still shrinking the overall series to a common value.

This work examines this parameterization (time-varying variance of log-volatility) for AR(1) log-volatility processes, integrated in a structural, order-invariant VAR model, as presented in Chan et al. (2024). The approach is compared with a more standard specification of the log-volatility processes, where an inverse-Gamma prior is used for the variance, which is assumed constant over time.

Both versions are examined, first, with a Monte Carlo simulation study with several different Data Generation Processes, and then with a real world application: a model of the global oil market which includes observations for the pandemic period.

The remaining of the thesis is structured as follows. Section 2 describes the SV-VAR framework, discusses identification conditions and presents the priors. Section 3 goes into some detail about the sampling algorithm. Section 4 discusses the issue of impact matrix scaling, and what is done to overcome it, with all implications for estimation. Section 5 describes the simulation study and discusses the main results. Section 6 presents the empirical application, including shock labeling, estimation results and a discussion of the role of the prior. Section 7 concludes.

2 Methodology

This section lays out the methodology used throughout the analysis, namely describing the VAR and log-volatility framework, explaining the conditions for identification of the impact matrix and describing the priors chosen.

2.1 Model description

The VAR is described by

$$\mathbf{y}_t = \mathbf{a} + \mathbf{A}_1\mathbf{y}_{t-1} + ... + \mathbf{A}_p\mathbf{y}_{t-p} + \mathbf{B}_0^{-1}\boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{D}_t) \tag{1}$$

where $\mathbf{a}$ is an $n \times 1$ vector of intercepts, $\mathbf{A}_1, ..., \mathbf{A}_p$ are $n \times n$ matrices of coefficients, $\mathbf{B}_0^{-1}$ is the (invertible) $n \times n$ impact matrix, $\boldsymbol{\varepsilon}_t$ is a vector of structural shocks, and $\mathbf{D}_t = diag(e^{h_{1,t}}, ..., e^{h_{n,t}})$ is its covariance matrix.

The processes $\boldsymbol{h}_i = (h_{i,1}, ..., h_{i,T})$, $i = 1, ..., n$ are the log-volatilities. Each follows an AR(1) process defined by

$$h_{i,t} = \phi_i h_{i,t-1} + \xi_{i,t}, \quad \xi_{i,t} \sim \mathcal{N}(0, \omega_{i,t}^2) \tag{2}$$

for $t = 1, ..., T$, where $|\phi_i| < 1$ and the initial condition $h_{i,0}$ is set to zero.

Further, let $\mathbf{u}_t = \mathbf{B}_0^{-1}\boldsymbol{\varepsilon}_t$ be the reduced-form shocks, and $\Sigma_t = \mathbf{B}_0^{-1}\mathbf{D}_t(\mathbf{B}_0^{-1})'$ its covariance matrix.

This corresponds to the OI-SV model from Chan et al. (2024) with the difference that the variance of the log-volatility error $\xi_{i,t}$ is dependent on t, and hence allowed to vary over time, and the initial condition is not drawn from the asymptotic distribution but fixed at zero.

2.2 Identification

The inclusion of Stochastic Volatility means that the impact matrix, or in this case its inverse, $\boldsymbol{B_0}$ is identified up to sign changes and row permutations.

Lütkepohl et al. (2025) show that, as long as the variances of the n structural shocks are different from each other for at least one time-point, then the rows of $\boldsymbol{B_0}$ are identified up to sign. The conditions are stated by the following Theorem (corresponding to Theorem 1 from Lütkepohl et al. (2025), here reproduced for convenience, with adapted notation):

Theorem 1 *Let* $\boldsymbol{\Sigma_t}$, $t = 0, 1, ...$ *be a sequence of positive definite* $n \times n$ *matrices, and* $\boldsymbol{D_t} = diag(\varsigma^2_{1,t}, ..., \varsigma^2_{n,t})$ *a sequence of* $n \times n$ *diagonal matrices with* $\boldsymbol{D_0} = \mathbf{1}\{n\}$. *Suppose there exists a nonsingular* $n \times n$ *matrix* $\boldsymbol{B_0}$ *such that*

$$\boldsymbol{\Sigma_t} = \boldsymbol{B_0^{-1}} \boldsymbol{D_t} (\boldsymbol{B_0^{-1}})', \quad t = 0, 1, ...$$

then, the i*-th row of* $\boldsymbol{B_0}$ *is unique up to sign if* $\boldsymbol{\varsigma}^2_i = (1, \varsigma^2_{i,1}, \varsigma^2_{i,2}, ...) \neq (1, \varsigma^2_{j,1}, \varsigma^2_{j,2}, ...) = \boldsymbol{\varsigma}^2_j$, $\forall j \in \{1, ..., n\} \backslash \{i\}$

It is easy to see that for the model defined by Equations 1-2, if the log-volatility of structural shock i, $\{\boldsymbol{h}_i\}$ differs from the log-volatility paths for other shocks at time point $t = t_d$ even if at all other points in time $e^{h_{i,t}} = e^{h_{j,t}}$, then

$$\boldsymbol{\varsigma}^2_i = (1, \varsigma^2_{i,1}, ..., \varsigma^2_{i,T}) = (e^{h_{i,0}}, e^{h_{i,1}}, ..., e^{h_{i,t_d}}, ..., e^{h_{i,T}}) \neq$$
$$\neq (e^{h_{j,0}}, e^{h_{j,1}}, ..., e^{h_{j,t_d}}, ..., e^{h_{j,T}}) = \boldsymbol{\varsigma}^2_j, \ \forall j \in \{1, ..., n\} \backslash \{i\}$$

and the i-th row of $\boldsymbol{B_0}$ is unique up to sign.

As it is extremely hard for the log-volatility processes of different structural shocks to be equal at all points in time, either theoretically or in practice, $\boldsymbol{B_0}$ should always be identified (again, up to sign and row permutations).

The connection between $\boldsymbol{\Sigma}_t, \boldsymbol{D}_t$ and $\boldsymbol{B_0}$ in the theorem is of note. It will be relevant for the discussion on scaling the $\boldsymbol{B_0}$ rows below.

2.3 Priors

The priors are set broadly as in Chan et al. (2024), the exception being the prior on $\boldsymbol{\omega^2_{i,t}}$.

Impact matrix

Let $\boldsymbol{b}_i$ be the i-th row of $\boldsymbol{B_0}$, so that $\boldsymbol{B_0'} = (\boldsymbol{b}_1, ..., \boldsymbol{b}_n)$. Chan et al. (2024) consider independent Gaussian priors on $\boldsymbol{b}_i$ with the i-th column of the identity $\boldsymbol{I}_n$ as the prior mean and $\boldsymbol{I}_n$ as well for the variance. These hyperparameters are chosen because they

represent an order invariant prior, in the sense that if the variables are transformed as $\tilde{\boldsymbol{y}}_t = \boldsymbol{P} \cdot \boldsymbol{y}_t$, for some permutation matrix $\boldsymbol{P}$, the prior mean of $\boldsymbol{PB_0P'}$ is also the identity and all prior variances of the elements are the same since $\boldsymbol{I}_n$ is diagonal. Setting the prior mean to the identity also makes column permutations less likely, avoiding the label-switching problem. In the present model the prior is, for each row $\boldsymbol{b}_i$

$$\boldsymbol{b}_i \sim \mathcal{N}(\boldsymbol{b}_{0,i}, v_b \boldsymbol{I}_n) \tag{3}$$

with the mean $\boldsymbol{b}_{0,i}$ kept as the i-th row of $\boldsymbol{I}_n$, but the prior variance is set at some multiple of the identity, as $v_b = 1$ is not informative enough in some cases. This preserves the order-invariant property and still makes label-switching less likely.

VAR coefficients

The prior for the VAR coefficients $\boldsymbol{a}, \boldsymbol{A}_1, ..., \boldsymbol{A}_p$ is the Minnesota-type Horseshoe prior proposed in Chan (2021). It is defined as follows. Let $\boldsymbol{\alpha}_i$ denote the coefficients of the i-th equation in the model, so $\boldsymbol{\alpha}_i = (\boldsymbol{a}_i, (\boldsymbol{A}_1)_{i1}, ..., (\boldsymbol{A}_1)_{in}, (\boldsymbol{A}_2)_{i1}, ..., (\boldsymbol{A}_p)_{in})$ where $\boldsymbol{a}_i$ is the intercept and $(\boldsymbol{A}_\ell)_{ij}$ is the $[ij]$ entry of of the $\boldsymbol{A}_\ell, \ell = 1, ..., p$ coefficient matrix.

For α_{ij}, the j-th entry of $\boldsymbol{\alpha}_i$, let $\kappa_{ij} = \kappa_1$ if it is a coefficient on an 'own lag' ($\alpha_{ij} = (\boldsymbol{A}_\ell)_{ii}, \ell = 1, ..., p$) and $\kappa_{ij} = \kappa_2$ if it is a coefficient on an 'other lag'. Let also C_{ij} be the constants defined as

$$C_{ij} = \begin{cases} \frac{1}{\ell^2}, & \text{for coefficient on } \ell\text{-th lag of variable } i, \\ \frac{s_i^2}{\ell^2 s_{i^*}^2}, & \text{for coefficient on } \ell\text{-th lag of variable } i^*, i^* \neq i. \end{cases}$$

where s_r^2 is the sample variance of the residuals from an AR(4) model for variable $r, r = 1, ..., n$. Then, the prior on each VAR coefficient $\alpha_{ij}, i = 1, ..., n, j = 2, ..., k$, $k = np + 1$ is a Horseshoe prior:

$$(\alpha_{ij} | \kappa_1, \kappa_2, \psi_{ij}) \sim \mathcal{N}(m_{ij}, \kappa_{ij}\psi_{ij}C_{ij})$$
$$\sqrt{\psi_{ij}} \sim \mathcal{C}^+(0,1) \ , \ \sqrt{\kappa_1}, \sqrt{\kappa_2} \sim \mathcal{C}^+(0,1)$$

where $\mathcal{C}^+(0,1)$ is the standard half-Cauchy distribution, κ_1, κ_2 are the (common to own or other lags) global variance components and ψ_{ij} is a local variance component. The

prior on the intercepts α_{i1} is simply $\mathcal{N}(m_{i1}, C_{i1}), C_{i1} = 100s_i^2$ [1].

Chan et al. (2024), like in Chan (2021), set $m_{ij} = 0$ for data in growth rates and $m_{ij} = 1$ for own lags when data is in levels. This is inspired by the usual Minnesota prior. Miranda-Agrippino and Ricco (2018), regarding this prior, refer the more general rule that the mean (m_{ij}) should be 0 for stationary data and 1 for coefficients on own lags when data is not stationary. The simulation study uses stationary data, so m_{ij} is set to zero there. In the empirical application, even though the variables are clearly not stationary, setting the prior mean as Miranda-Agrippino and Ricco (2018) suggest results in unstable coefficient estimates. Setting all $m_{ij} = 0, \forall i = 1, ..., n, j = 1, ..., k$ works much better, with both versions of the model. So, this is the prior mean used for all estimation.

Stochastic Volatility

For the stochastic volatility autoregressive parameter ϕ_i, the prior is a truncated Normal

$$\phi_i \sim \mathcal{N}(\phi_{0i}, V_{\phi_i}) \cdot \mathbf{1}\{|\phi| < 1\} \tag{4}$$

For the prior on $\boldsymbol{\omega}_i^2$, which is at the center of this work, 2 versions are tested.

For a flexible model, with time-varying variance of log-volatility (ω_{it}^2), a Horseshoe prior is used:

$$\omega_{it}^2 = \tau_i \lambda_{it},$$
$$\sqrt{\tau_i} \sim \mathcal{C}^+(0,1) \ , \ \sqrt{\lambda_{it}} \sim \mathcal{C}^+(0,1)$$

The standard approach considers $\forall t : \omega_{it}^2 = \omega_i^2$ and places a tightly parameterized inverse-Gamma prior on ω_i^2. This is the approach followed in Chan et al. (2024), who set the prior $\mathcal{IG}(\nu_i, S_i)$. A version of the model using this prior, with $\nu_i = 5$ and $S_i = 0.04$, is estimated in the simulation studies and empirical application for comparison with the time-varying ω_{it}^2 case. These model versions are referred to as the HS and IG model, respectively, referencing the prior which distinguishes them.

[1] meaning the shrinkage on the intercepts is very weak

3 MCMC algorithm

This section lays out the posterior sampler steps used for estimation of the model. The sampler as a whole follows a similar structure to the one in Chan et al. (2024). Only the steps that are significantly different from their counterparts in Chan et al. (2024) are explained in more detail in the section. For a detailed explanation of the remaining steps, see Appendix A.

Regarding notation, let $\boldsymbol{\psi}_i = (\psi_{i,1}, ..., \psi_{i,k})'$ denote the local variance component of $\boldsymbol{\alpha}_i$, with $\boldsymbol{z_{\psi_i}} = (z_{\psi_{i,1}}, ..., z_{\pi_{i,k}})'$ the latent variables corresponding to $\boldsymbol{\phi}_i$. Let also $\boldsymbol{\kappa} = (\kappa_1, \kappa_2)'$ and $\boldsymbol{z_{\kappa_i}} = (z_{\kappa_1}, z_{\kappa_2})'$ its latent variables.

The posterior draws are obtained by sequentially sampling from the conditional distributions of the parameters as follows:

1. Sample $\mathbf{B}_0$ from $p(\mathbf{B}_0 \mid \mathbf{y}, \boldsymbol{\alpha}, \mathbf{h})$
2. Sample $\boldsymbol{\alpha}$ from $p(\boldsymbol{\alpha} \mid \mathbf{y}, \mathbf{B}_0, \mathbf{h}, \boldsymbol{\psi}, \boldsymbol{\kappa})$
3. Sample $\mathbf{h}$ from $\prod_{i=1}^{n} p(h_{i,\cdot} \mid \mathbf{y}, \mathbf{B}_0, \boldsymbol{\alpha}, \boldsymbol{\phi}, \boldsymbol{\omega}_t^2)$
4. Sample ω from $\prod_{i=1}^{n} p(\omega_i^2 \mid h_{i,\cdot}, \phi_i)$ or $\prod_{i=1}^{n} \prod_{t=1}^{T} p(\omega_{it}^2 \mid h_{i,\cdot}, \phi_i)$
5. Sample $\boldsymbol{\phi}$ from $\prod_{i=1}^{n} p(\phi_i \mid h_{i,\cdot}, \omega_{it}^2)$
6. Sample $\boldsymbol{\psi}$ from $\prod_{i=1}^{n} \prod_{j=2}^{k} p(\psi_{i,j} \mid \alpha_{i,j}, \boldsymbol{\kappa}, z_{\psi_{i,j}})$
7. Sample $\boldsymbol{z_\psi}$ from $\prod_{i=1}^{n} \prod_{j=2}^{k} p(z_{\psi_{i,j}} \mid \psi_{i,j})$
8. Sample $\boldsymbol{\kappa}$ from $\prod_{l=1}^{2} p(\kappa_l \mid \boldsymbol{\alpha}, \boldsymbol{\psi}, z_{\kappa_l})$
9. Sample $\boldsymbol{z_\kappa}$ from $\prod_{l=1}^{2} p(z_{\kappa_l} \mid \kappa_l)$

The steps that are different in some way from the corresponding step in Chan et al. (2024) are Steps 1, 3, 4 and 5. These parameters are sampled as follows:

Step 1. Each row $\mathbf{b}_i$ of $\boldsymbol{B_0}$ is sampled separately, given all other rows.

Rewriting 1 as

$$(\mathbf{Y} - \mathbf{XA})\boldsymbol{B_0}' = \mathbf{E} \tag{5}$$

where $\boldsymbol{Y}$ is the matrix of stacked $\boldsymbol{y}_t$ vectors, so $\boldsymbol{Y} = [\boldsymbol{y}_1\ \boldsymbol{y}_2\ ...\ \boldsymbol{y}_T]'$, $\boldsymbol{X}$ is the matrix of lagged variables, $(T \times k)$, $k = np + 1$, $\boldsymbol{A} = (\boldsymbol{a}, \boldsymbol{A}_1, ..., \boldsymbol{A}_p)'$ is the matrix of VAR coefficients and $\boldsymbol{E}$ is the matrix of errors. Then, for $i = 1, ..., n$

$$(\mathbf{Y} - \mathbf{XA})\mathbf{b}_i = (\mathbf{U})\mathbf{b}_i = \mathbf{E}_i, \quad \boldsymbol{E}_i \sim \mathcal{N}(\mathbf{0}_T, \boldsymbol{\Omega}_{\boldsymbol{h}_i}) \tag{6}$$

where $\boldsymbol{E}_i$ is the i-th column of $\boldsymbol{E}$ and $\boldsymbol{\Omega}_{\boldsymbol{h}_i} = diag(e^{h_{i,1}}, ..., e^{h_{i,T}})$. The full conditional distribution of $\mathbf{b}_i$ is given by

$$\begin{aligned} p(\mathbf{b}_i \mid \mathbf{y}, \boldsymbol{\alpha}, \mathbf{b}_{-i}, \mathbf{h}_i, \cdot) &\propto |\det \mathbf{B}_0|^T \, e^{-\frac{1}{2}\mathbf{b}_i' \boldsymbol{U}' \boldsymbol{\Omega}_{\boldsymbol{h}_i}^{-1} \boldsymbol{U} \mathbf{b}_i} \quad \times e^{-\frac{1}{2} v_b (\mathbf{b}_i - \mathbf{b}_{0i})'(\mathbf{b}_i - \mathbf{b}_{0i})} \\ &\propto |\det \mathbf{B}_0|^T \, e^{-\frac{1}{2}(\mathbf{b}_i - \hat{\mathbf{b}}_i)' \mathbf{K}_{\mathbf{b}_i} (\mathbf{b}_i - \hat{\mathbf{b}}_i)} \end{aligned}$$

with $\mathbf{K}_{\mathbf{b}_i} = \frac{1}{v_b} \cdot \boldsymbol{I}_n + \boldsymbol{U}' \boldsymbol{\Omega}_{\boldsymbol{h}_i} \boldsymbol{U}$ and $\hat{\mathbf{b}}_i = \mathbf{K}_{\mathbf{b}_i}^{-1} \cdot \frac{1}{v_b} \cdot \boldsymbol{I}_n \mathbf{b}_{0i}$.

This is a non-standard posterior, hence direct sampling is not feasible. The sampling approach follows the one from Villani (2009), a generalization of the algorithm from Waggoner and Zha (2003) to a non-zero $\hat{\mathbf{b}}_i$. It can be shown that $\mathbf{b}_i$ has the same distribution as a linear transformation of an n-vector of one absolute normal random variable and $n - 1$ normally distributed ones.

Formally, let $\boldsymbol{G}_i$ be the Cholesky factor of $T^{-1}\mathbf{K}_{\mathbf{b}_i}$, $\mathbf{v}_1 = \mathbf{G}_i^{-1}(\mathbf{B}_{0,-i}')^{\perp} / \|\mathbf{G}_i^{-1}(\mathbf{B}_{0,-i}')^{\perp}\|$ and $(\mathbf{v}_2, \ldots, \mathbf{v}_n) = \mathbf{v}_1^{\perp}$. From Chan et al. (2024),

$$(\mathbf{b}_i \mid \mathbf{y}, \boldsymbol{\alpha}, \mathbf{b}_{-i}, \mathbf{h}_i, \cdot) \overset{d}{=} (\mathbf{G}_i')^{-1} \sum_{j=1}^{n} \xi_j \mathbf{v}_j, \quad \xi_1 \sim \mathcal{N}(\widehat{\xi}_1, T^{-1}), \quad \xi_j \sim \mathcal{N}(\widehat{\xi}_j, T^{-1}), j = 2, \ldots, n$$

where $\widehat{\xi}_j = \mathbf{b}_i' \mathbf{C}_i \mathbf{v}_j$. Chan et al. (2024) offers a proof of this claim.

For sampling from the absolute normal distribution, the 2-component normal mixture approximation from Villani (2009) is used. Further, during sampling the sign of the i-th element of $\mathbf{b}_i$ is fixed to be positive, so that the diagonal of $\boldsymbol{B_0}$ is positive, by multiplying the draw of $\mathbf{b}_i$ by the sign of the i-th element. This is a normalization step already present in the Chan et al. (2024) sampler, to avoid the sign switching issue. It is unrelated to the normalization described in Section 4.1 below, which is done post-sampling, specifically to deal with the scaling issue described there.

Step 3. The log-volatility vector $\mathbf{h}_i$ of each variable $i = 1, ..., n$ is sampled with the help of both the auxiliary mixture sampler of Kim et al. (1998) and the band matrix precision method of Chan and Jeliazkov (2009) for linear Gaussian state space models.

From 6, we have $\boldsymbol{E}_i \sim \mathcal{N}(\mathbf{0}_T, \boldsymbol{\Omega}_{\boldsymbol{h}_i})$, meaning each $E_{i,t}$ follows a $\mathcal{N}(0, e^{h_{i,t}})$ distribution, or equivalently

$$E_{i,t} = e^{\frac{h_{i,t}}{2}} \cdot u^*_{i,t}, \quad u^*_{i,t} \sim \mathcal{N}(0, 1)$$

By a simple transformation, the state equation 2 becomes linear in $h_{i,t}$

$$y^*_{i,t} = \log(E^2_{i,t}) = h_{i,t} + \log({u^*_{i,t}}^2)$$

The error in this equation is no longer Gaussian, but log-chi-square. Let it be defined $\epsilon^*_{i,t} = \log({u^*_{i,t}}^2)$. Its distribution can be approximated by the well-known 7-component mixture of normals from Kim et al. (1998), written in terms of a component indicator variable s_t

$$\epsilon^*_{i,t}|s_t = j \sim \mathcal{N}(d_t = \mu_j - 1.2704, \sigma^2_{s_t} = v^2_j), \quad \Pr(s_t = j) = q_j, \quad j = 1, ..., 7$$

with $\mu_1, ..., \mu_7$, $v^2_1, ..., v^2_7$ and $q_1, ..., q_7$ known values (for details see Kim et al. (1998)). Stacking, $(\boldsymbol{\epsilon^*}|\mathbf{s}) \sim \mathcal{N}(\mathbf{d}, \Omega^*)$, with $\Omega^* = diag(\sigma^2_{s_1}, ..., \sigma^2_{s_T})$, and it follows that $(\boldsymbol{y^*}|\mathbf{s}, \mathbf{h}) \sim \mathcal{N}(\mathbf{h} + \mathbf{d}, \Omega^*)$.

Now stacking the state equations, we can rewrite them as

$$\boldsymbol{H}_{\boldsymbol{\phi}}\mathbf{h}_i = \boldsymbol{\xi}_i, \quad \boldsymbol{\xi}_i = (\xi_{i,1}, ..., \xi_{i,T})' \sim \mathcal{N}(0, \mathbf{S}_i)$$

where

$$\boldsymbol{H}_{\boldsymbol{\phi_i}} = \begin{bmatrix} 1 & & & & \\ -\phi_i & 1 & & & \\ & -\phi_i & 1 & & \\ & & \ddots & \ddots & \\ & & & -\phi_i & 1 \end{bmatrix}, \quad \mathbf{S}_i = \begin{bmatrix} \frac{\omega^2_{i,1}}{1-{\phi_i}^2} & & & & \\ & \omega^2_{i,2} & & & \\ & & \omega^2_{i,3} & & \\ & & & \ddots & \\ & & & & \omega^2_{i,T} \end{bmatrix}$$

By standard normal regression results, this implies that $(\mathbf{h}_i|\boldsymbol{\phi_i}, \boldsymbol{\omega}_i, \Omega^*) \sim \mathcal{N}(0, \mathbf{P}_i^{-1})$, where the precision matrix $\mathbf{P}_i = \boldsymbol{H}'_{\boldsymbol{\phi_i}} \mathbf{S}_i^{-1} \boldsymbol{H}_{\boldsymbol{\phi_i}}$ is tridiagonal. As $p(\mathbf{h}_i|\mathbf{y}^*, \mathbf{s}, \phi_i, \omega_i) \propto p(\mathbf{y}^*|\mathbf{s}, \mathbf{h}_i) p(\mathbf{h}_i|\phi_i, \omega_i)$ and both this likelihood and prior are normal, it is trivial to show that

$$(\mathbf{h}_i|\mathbf{y}^*, \mathbf{s}, \phi_i, \omega_i) \sim \mathcal{N}(\hat{\mathbf{h}}_\mathbf{i}, \mathbf{K}_i^{-1})$$

with $\mathbf{K}_i = \mathbf{P}_i + \Omega^{*-1}$ and $\hat{\mathbf{h}}_\mathbf{i} = \mathbf{K}_i^{-1} \Omega^{*-1} (\mathbf{y}^* - \mathbf{d})$. It is possible to efficiently sample from this high-dimensional Gaussian distribution using the precision sampler from Chan and Jeliazkov (2009), which exploits the special structure of $\mathbf{K}_i^{-1}$. Chan and Hsiao (2014) discuss in detail the use of this sampler for the case where the mean of the distribution of $\mathbf{h}_{\mathbf{i},\mathbf{1}}$ is not zero and $\omega_{i,t}^2 = \omega_i \forall t = 1, ..., T$, and argue its efficiency compared to Kalman filter-based algorithms.

Step 4. This step depends on the model being sampled.

In the case of a time-constant $\omega_{i,t} = \omega_i, \forall t = 1, ..., T$ with an inverse-Gamma prior $\mathcal{IG}(\nu_i, S_\omega)$, the posterior is standard (see Chan and Hsiao (2014)): it is inverse-Gamma as well. It is straightforward to independently draw each ω_i^2 from its conditional distribution $\mathcal{IG}(\nu_i + \frac{T}{2}, \tilde{S}_\omega)$, where $\tilde{S}_\omega = S_\omega + \left((1 - \phi_i^2) h_{i,1}^2 + \sum_{t=2}^{T} (h_{i,t} - \phi_i h_{i,t-1})^2 \right) \frac{1}{2}$.

In the case of time-varying $\omega_{i,t}$ with a Horseshoe prior defined by $\omega_{i,t}^2 = \tau_i \lambda_{t,i}$, $\sqrt{\tau_i} \sim \mathcal{C}^+(0, 1)$, $\sqrt{\lambda_{t,i}} \sim \mathcal{C}^+(0, 1)$, we can use the scale mixture representation of the Half-Cauchy distribution: if X and w are random variables such that $(X^2|w) \sim \mathcal{IG}(\frac{1}{2}, \frac{1}{w})$ and $w \sim \mathcal{IG}(\frac{1}{2}, 1)$, then $X \sim \mathcal{C}^+(0, 1)$.

Thus, defining the latent variables ν_{τ_i} and $\nu_{\lambda_{i,t}}$, it is straightforward to derive the posteriors to sample from:

$$\begin{aligned}
(\tau_i|\mathbf{h}, \nu_{\tau_i}, \lambda_{i,t}) &\sim \mathcal{IG}\left(\frac{T+1}{2}, \frac{1}{\nu_{\tau_i}} + \frac{1}{2} \sum_{t=1}^{T} \frac{(h_{i,t} - \phi_i h_{i,t-1})^2}{\lambda_{i,t}} \right) \\
(\lambda_{i,t}|\mathbf{h}, \nu_{\lambda_{i,t}}, \tau_i) &\sim \mathcal{IG}\left(1, \frac{1}{\nu_{\lambda_{i,t}}} + \frac{1}{2} \frac{(h_{i,t} - \phi_i h_{i,t-1})^2}{\tau_i} \right) \\
(\nu_{\tau_i}|\tau_i) &\sim \mathcal{IG}\left(1, 1 + \frac{1}{\tau_i} \right) \\
(\nu_{\lambda_{i,t}}|\lambda_{i,t}) &\sim \mathcal{IG}\left(1, 1 + \frac{1}{\lambda_{i,t}} \right)
\end{aligned}$$

Each τ_i, $\lambda_{i,t}$, ν_{τ_i} and $\nu_{\lambda_{i,t}}$ can be sampled independently.

Step 5. This step is adapted from Chan and Hsiao (2014). The posterior of ϕ_i is given by

$$p(\phi_h \mid \mathbf{h}_i, \mathbf{h}_0 = 0, \omega_{it}^2) \propto p(\phi_i) g(\phi_i) e^{-\frac{1}{2}\sum_{t=2}^{T} \frac{(h_{it} - \phi_i h_{i,t-1})^2}{\omega_{it}^2}}$$

with $g(\phi_i) = \sqrt{(1-\phi_i^2)} \exp\{-\frac{1}{2\omega_{i,1}^2}(1-\phi_i^2)h_{i1}^2\}$ and $p(\phi_i)$ the truncated normal prior from 4. This density is non-standard, so it is sampled through an independence-chain Metropolis-Hastings step.

The proposal distribution used is the truncated normal $\mathcal{N}(\hat{\phi}_i, D_{\phi_i}) \cdot 1(|\phi_i| < 1)$, with $D_{\phi_i} = (V_{\phi_i}^{-1} + \sum_{t=1}^{T-1} \frac{h_{it}^2}{\omega_{it}^2})^{-1}$ and $\hat{\phi}_h = D_{\phi_i}(V_{\phi_i}^{-1}\phi_{i0} + \sum_{t=2}^{T} \frac{(h_{it}h_{i,t-1})}{\omega_{it}^2})$. Given the current draw ϕ_i, a proposal ϕ_i^* is accepted with probability $\min(1, g(\phi_i^*)/g(\phi_i))$ (otherwise rejected and the Markov chain stays at the current draw).

In the estimation, a very tight prior variance is chosen for ϕ_i, $V_{\phi_i} = 10^{-18}$, around a mean of 0.95 for each $\phi_i, i = 1, ..., n$. This was found to be necessary during simulations for convergence of the parameters of interest ($\boldsymbol{B_0}$, $\boldsymbol{\alpha}$ and $\boldsymbol{h}_t$).

4 Challenges

The employed framework comes with some particular challenges that must be dealt with before continuing to the main analysis. This Section illustrates the problems, explains how they arise, and discusses how to overcome them.

4.1 Challenges: The Scaling issue

In a stochastic volatility-identified SVAR model, the impact matrix is famously identified only up to sign and column/row (depending on parameterization) permutations (Bertsche and Braun (2020), Lütkepohl et al. (2025)). It is also a concern in some works that the columns/rows are potentially incorrectly scaled. This issue is noted, for example, by Brunnermeier et al. (2021), who warn that if an impact matrix is rescaled by

some factor and so is the structural volatility matrix, this does not affect the behavior of the data.

At a first glance, this seems like it could contradict Theorem 1. This is not the case: the identification Theorem can be valid and the scaling issue still be present. Theorem 1 ensures that the rows of $\mathbf{B}_0$ are unique up up to sign if the respective variance paths are different from one another for at least one point in time. The theorem establishes a connection between the variance of structural shocks $\boldsymbol{\varepsilon}_t$ and the impact matrix. It does not, however, ensure that the correct variance and hence the correct $\mathbf{B}_0$ are computed. It is possible then for an incorrect n-th impact matrix column and respective incorrect structural shock's variance process, $e^{h_{n,t}}$, to be estimated together, and this would not contradict the theorem. In fact, if a row of $\mathbf{B}_0$ is estimated multiplied by some factor, the corresponding structural variance is multiplied by the same quantity (squared), and nothing else is different, the data would have the exact same behavior.

To prove this, assume that each row $\mathbf{b}_i$ of $\mathbf{B}_0$ is estimated scaled by a constant c_i. Meaning, the estimate of the impact matrix is actually $\tilde{\mathbf{B}}_0 = \mathbf{C} \cdot \mathbf{B_0}$, with $\mathbf{C} = diag(c_1, ..., c_n)$. Because the model is as described by Equation 1, it is apparent that the data has the same behavior under the scaled impact matrix, provided that the structural errors are also scaled ($\tilde{\boldsymbol{\varepsilon}}_t = \boldsymbol{C} \cdot \boldsymbol{\varepsilon}_t$). Defining $\tilde{\boldsymbol{u}}_t$ as the reduced-form errors computed with $\tilde{\mathbf{B}}_0$ and $\tilde{\boldsymbol{\varepsilon}}_t$, we have

$$\tilde{\boldsymbol{u}}_t = (\mathbf{CB_0})^{-1}\tilde{\boldsymbol{\varepsilon}}_t = \mathbf{B}_0^{-1}\mathbf{C}^{-1}\boldsymbol{C}\boldsymbol{\varepsilon}_t = \mathbf{B}_0^{-1}\boldsymbol{\varepsilon}_t = \mathbf{u}_t \tag{7}$$

which is the vector of reduced-form errors of the correct model. Note also that the covariance matrix of $\tilde{\boldsymbol{\varepsilon}}_t$ is $\tilde{\boldsymbol{D}}_t = \boldsymbol{C}\boldsymbol{D}_t\boldsymbol{C}'$.

This issue is present in the estimation of this model. The source here is simple, and a standard concern in Bayesian estimation. It has to do with the prior mean and variance for $\mathbf{B}_0$. As stated in Section 2.3, the prior for each row i of $\mathbf{B}_0$ is $\mathcal{N}(\mathbf{b_0}_i, v_b \cdot \mathbf{I}_n)$, where $\mathbf{b_0}_i$ is the i-th row of the identity matrix and v_b a constant. Naturally, tightening the prior variance of $\mathbf{B}_0$ moves the estimate closer to the prior mean, which is the identity. As the prior is defined row by row, each entry of the estimated matrix is 'pulled' away from its true value by the same multiplicative factor, resulting in a scaling of the estimates by a certain constant c_i.

It should be noted, the cause is not necessarily the prior variance in all the works cited

above, nor is it claimed so here. In the current model, however, this seems to be the source of the incorrect scaling, as demonstrated below.

The problem can be illustrated with the following example: Let a process be generated according to Equations 1-2, with $\boldsymbol{a} = (2, 8, 4)'$ and

$$\mathbf{A}_1 = \begin{bmatrix} 0.2 & 0.05 & 0 \\ -0.12 & 0.3 & 0.1 \\ 0.05 & -0.05 & 0.25 \end{bmatrix}, \mathbf{A}_2 = \begin{bmatrix} 0.1 & 0.2 & 0 \\ -0.15 & 0.3 & 0.05 \\ 0 & 0.1 & 0.2 \end{bmatrix}, \mathbf{B_0} = \begin{bmatrix} 3 & 1.2 & -0.8 \\ 0.9 & 2 & 1.1 \\ -0.6 & 0.7 & 2.5 \end{bmatrix}$$

with $T = 400$, $\phi_i = 0.95$ and $\omega_{i,t}^2 = 0.05, \forall t = 1, ..., T$ but with a 2 unit 'jump' in h_{it} at $t = 320$ [2].

Estimating the model with the Horseshoe prior version, $v_b = 0.5$, with 100000 post-burnin iterations (thinned by 100), the MCMC draws converge [3], but to different values than $\boldsymbol{B_0}$. The median of draws (estimates with the mean are similar) is

$$\hat{\boldsymbol{B}}_{\mathbf{0}} = \begin{bmatrix} 2.72 & 1.09 & -0.69 \\ 0.80 & 1.84 & 1.02 \\ -0.44 & 0.57 & 1.98 \end{bmatrix} \approx \begin{bmatrix} 0.9 & 0 & 0 \\ 0 & 0.9 & 0 \\ 0 & 0 & 0.8 \end{bmatrix} \begin{bmatrix} 3 & 1.2 & -0.8 \\ 0.9 & 2 & 1.1 \\ -0.6 & 0.7 & 2.5 \end{bmatrix}$$

Naturally, this is only and approximation, but this happens consistently across datasets. It can be illustrated with a small simulation exercise: Generate $R = 300$ processes $\boldsymbol{h}_i, i = 1, 2, 3$ and respective $\boldsymbol{\varepsilon}_t$ and $\boldsymbol{y}_t$ according to Eqs. 1-2. On each dataset $r = 1, ..., R$ estimate the model as described above, with 3 different values for v_b, 1, 0.5 and 0.1.

Figure 1 shows the distribution of the estimated entries of $\boldsymbol{B_0}$.

The tighter v_b is set, the closer the estimates are, on average, to their respective entry of the prior mean (one for diagonal entries, zero for others).

Furthermore, this has no impact on the estimated autoregressive coefficients: the distribution of estimates for $\boldsymbol{a}$ and $\boldsymbol{A}_i$ are the same for every v_b (reported in Appendix B).

[2] This corresponds to the DGP2 equations. More details in Section 5

[3] see Appendix for trace plots

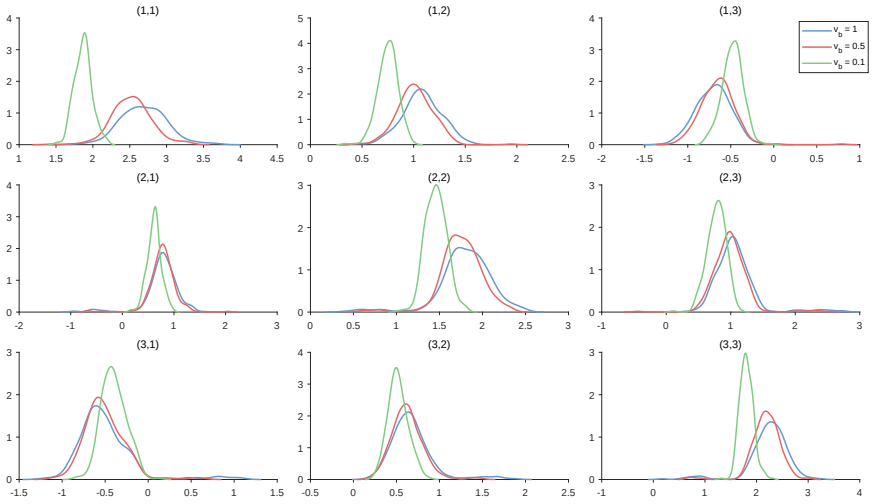


Figure 1: Densities of estimated B0 entries with different v_b across $R = 300$ datasets. The estimate is the median of draws. The plots are organized like the entries they correspond to, so the first plot is the $(1, 1)$ entry of $\boldsymbol{B_0}$, etc.

This happens because the scaling of the matrix rows is 'compensated' by a similar scaling of the structural error variance processes , so the model estimates $\tilde{\boldsymbol{D}}_t$ instead of $\boldsymbol{D}_t$. This $\tilde{\boldsymbol{D}}_t$ is given by $\boldsymbol{C}\boldsymbol{D}_t\boldsymbol{C}' = diag(c_1^2 e^{h_{1,t}}, ..., c_n^2 e^{h_{n,t}}) = diag(e^{h_{1,t}+2\ln c_1}, ..., e^{h_{n,t}+2\ln c_n})$. This implies the log-volatilies estimated are $\tilde{h}_{it} = h_{it} + 2\ln c_i, i = 1, ..., n$. The closer the $\boldsymbol{B_0}$ estimates are pulled towards the identity, the farther the estimated h_{it} processes are pushed away from zero. Figure 2 illustrates this effect.

As stated before, this problem is not new. The solutions tend to focus on restricting the structural variance processes. The restriction $\mathbf{E}(\boldsymbol{h}_{it}) = 0$ is routinely set to identify the scale of shocks, but often found to not be enough. Brunnermeier et al. (2021) further restrict the sample mean of the structural variances to be equal to 1 for each variable, i.e., $\frac{1}{T}\sum_{t=1}^{T} e^{h_{i,t}} = 1, \forall i = 1, ..., n$. Bertsche and Braun (2020) do something similar, but restricting the log-volatility sample mean instead: $\frac{1}{T}\sum_{t=1}^{T} h_{i,t} = 0, \forall i = 1, ..., n$. Meanwhile Lewis (2021) also stresses that the level of process h_{it} should be set to zero.

These restrictions are placed because the scale of the shocks or the impact matrix itself is of interest to the researchers. In the present case, however, it is precisely of interest to compare the estimated variance of shocks, and constraining its sample moments could distort the conclusions. Therefore, the log-volatilities are left (relatively) unconstrained, meaning the sampler can only obtain estimates for $\tilde{\boldsymbol{B}}_{\boldsymbol{0}}$ and $\tilde{\boldsymbol{\varepsilon}}_t$.

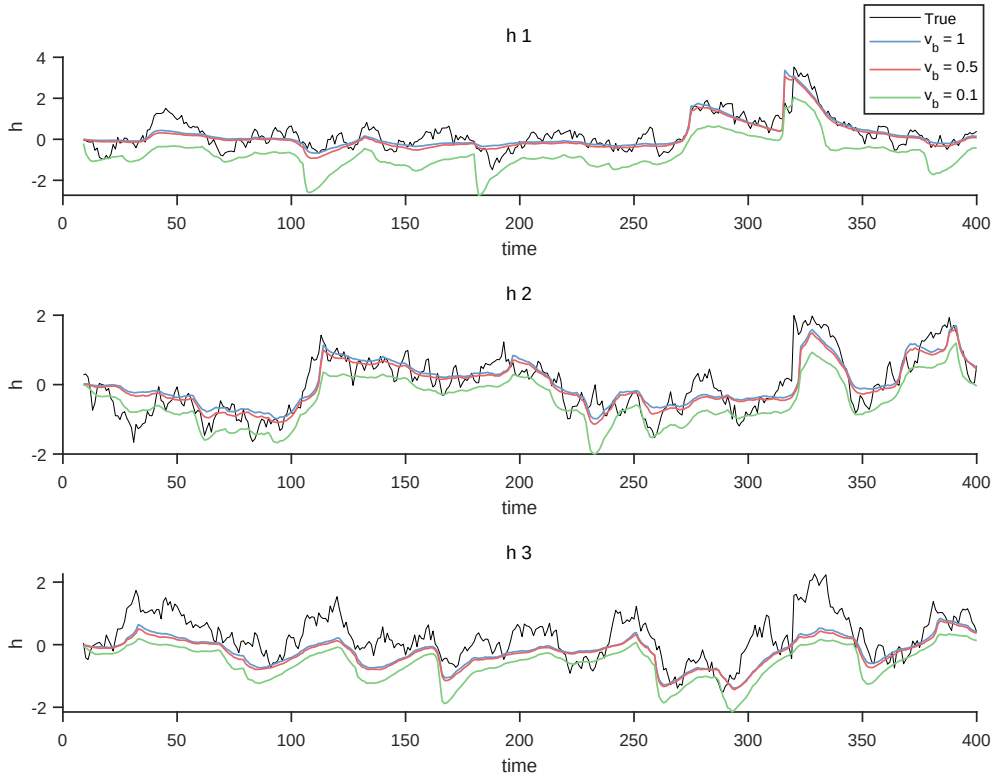


Figure 2: Estimated h with different v_b for an exemplary dataset r out of the $R = 300$.

There is, however, some information still intact. Since each row is scaled by the same constant, the size of each element relative to the diagonal entry of its row is preserved. So, normalizing each row of $\boldsymbol{B_0}$ so that entries are shown as ratios to the corresponding diagonal entry yields a matrix, $\boldsymbol{B_0^{norm}}$, for which the estimates must be consistent across choices of v_b. The normalization is done as follows:

1. N draws are obtained from the sampler, $\tilde{\boldsymbol{B_0}}^{(d)}$ being the d-th draw of $\tilde{\boldsymbol{B_0}}$
2. Each row of each sampled $\tilde{\boldsymbol{B_0}}^{(d)}$, $\tilde{\boldsymbol{b_i}}^{(d)}$ is divided by the mean of all draws of the diagonal entry in that row, $\bar{\tilde{b}_{ii}} = \frac{1}{N}\sum_{d=1}^{N} \tilde{b}_{ii}^{(d)}$

This brings the estimates of the diagonal entries to 1 when using the mean as point estimate, close to 1 when using the median.

Figure 3 shows the distributions of the normalized estimates for the simulation described

above.

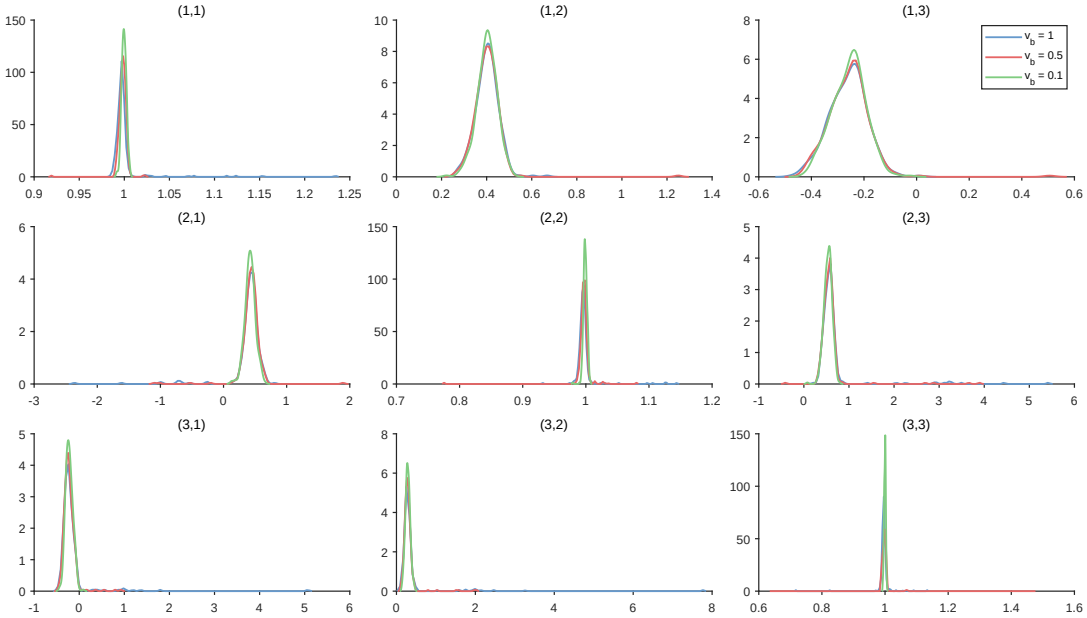


Figure 3: Densities of estimated $\boldsymbol{B}_{\mathbf{0}}^{norm}$ entries with different v_b across $R = 300$ datasets. The estimate is the median of draws. The plots are organized like the entries they correspond to.

These results hold for other instances of the $\boldsymbol{B_0}$ as well. Densities of estimated entries of the (inverse) impact matrix, normalized and non-normalized, for a similar study with an alternative true $\boldsymbol{B_0}$ are reported in the Appendix.

To help with formal definition of some results from here onward, it is helpful to see the normalized $\boldsymbol{B_0}$ as a multiplication of matrices. Let $diagMat(\boldsymbol{M})$ define the matrix which has the same diagonal as $\boldsymbol{M}$ and zeros in the other entries. Further, let $\boldsymbol{Z}$ be the inverse of $diagMat(\boldsymbol{B_0})$. Then,

$$\tilde{\boldsymbol{B}_{\mathbf{0}}}^{norm} = diagMat(\tilde{\boldsymbol{B}_{\mathbf{0}}})^{-1} \cdot \tilde{\boldsymbol{B}_{\mathbf{0}}} = diagMat(\boldsymbol{C} \cdot \boldsymbol{B_0})^{-1}\boldsymbol{C} \cdot \boldsymbol{B_0} \tag{8}$$

$$= diagMat(\boldsymbol{C})^{-1}\boldsymbol{Z}\boldsymbol{C} \cdot \boldsymbol{B_0} = \boldsymbol{Z}\boldsymbol{C}^{-1}\boldsymbol{C}\boldsymbol{B_0} = \boldsymbol{Z}\boldsymbol{B_0} \tag{9}$$

The reduced-form variances $\boldsymbol{\Sigma}_t = \boldsymbol{B}_{\mathbf{0}}^{-1}\boldsymbol{D}_t(\boldsymbol{B}_{\mathbf{0}}^{-1})'$ should still be computed with the non-

normalized estimated matrix, as

$$\begin{aligned}(\boldsymbol{B}_{\mathbf{0}}^{norm})^{-1}\tilde{\boldsymbol{D}}_t((\boldsymbol{B}_{\mathbf{0}}^{norm})^{-1})' &= (\boldsymbol{Z}\boldsymbol{B}_{\mathbf{0}})^{-1}\tilde{\boldsymbol{D}}_t((\boldsymbol{Z}\boldsymbol{B}_{\mathbf{0}})^{-1})' \\ &= \boldsymbol{B}_{\mathbf{0}}^{-1}\boldsymbol{Z}^{-1}\boldsymbol{C}\boldsymbol{D}_t\boldsymbol{C}(\boldsymbol{Z}^{-1})'(\boldsymbol{B}_{\mathbf{0}}^{-1})' \neq \boldsymbol{\Sigma}_t\end{aligned}$$

It is clear then that the estimated $\boldsymbol{B}_{\mathbf{0}}$, $\boldsymbol{h}_t$ and shock variances in the studied model are scaled by some unknown factor. It is possible to normalize the impact matrix draws, standardizing the result across different prior variances, to still retrieve the true entries of $\boldsymbol{B}_{\mathbf{0}}$ as a ratio to the diagonal. This must be handled with care, as it changes how information is retrieved from the model, namely covariances matrices and VAR tools. The latter are discussed in the next section.

4.2 Challenge: VAR tools

The fact that the structural shocks' covariance matrix, and so also the reduced-form shock variance matrix, is time dependent has some implications for standard VAR tools which are of interest to examine, particularly in the empirical application. The scaling issue discussed above and subsequent normalization of $\boldsymbol{B}_{\mathbf{0}}$ also influence how those tools should be computed in order to ensure a fair comparison between the HS and IG models. So, this section briefly discusses how those tools, namely Impulse Response Functions (IRFs), Forecast Error Variance Decompositions (FEVDs) and Historical Decompositions (HDs), should be computed in this particular setting.

The definition of IRFs, FEVDs, and HDs followed here is taken from Kilian and Lütkepohl (2017).

IRF: For the model defined by 1, the $n \times n$ matrix of impulse responses is $\boldsymbol{\Theta}_\tau$, for each horizon $\tau = 0, 1, ..., H$. The jk-th entry of $\boldsymbol{\Theta}_\tau$, $\theta_{jk,\tau}$, represents the response of variable j at time $t+\tau$ ($y_{j,t+\tau}$) to the structural shock k at time t ($\varepsilon_{k,t}$). These impulse responses are computed as $\boldsymbol{\Theta}_\tau = \boldsymbol{\Phi}_\tau \cdot \boldsymbol{B}_{\mathbf{0}}^{-1}$, with $\boldsymbol{\Phi}_\tau$ an $n \times n$ matrix representing the responses of

$\boldsymbol{y}_{t+\tau}$ to reduced-form shocks $\boldsymbol{u}_t$. This matrix is computed as $\boldsymbol{J}\boldsymbol{A}^{\tau}\boldsymbol{J}'$, with

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \cdots & \mathbf{A}_{p-1} & \mathbf{A}_p \\ \mathbf{I}_n & 0 & \cdots & 0 & 0 \\ 0 & \mathbf{I}_n & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \mathbf{I}_n & 0 \end{bmatrix}, \quad \mathbf{J} = \begin{bmatrix} \mathbf{I}_n & \mathbf{0}_{n\times n(p-1)} \end{bmatrix}$$

The time-varying $\boldsymbol{\Sigma}_t$ does not alter the way the IRFs are computed, as the equivalence comes from the MA representation of $\boldsymbol{y}_t$, $\boldsymbol{y}_t = \sum_{i=0}^{\infty} \boldsymbol{\Phi}_i \boldsymbol{u}_{t-i} = \sum_{i=0}^{\infty} \boldsymbol{\Phi}_i \boldsymbol{B_0}^{-1} \boldsymbol{B_0} \boldsymbol{u}_{t-i} = \sum_{i=0}^{\infty} \boldsymbol{\Theta}_i \boldsymbol{\varepsilon}_{t-i}$, which does not change with the variance of $\boldsymbol{u}_t$.

Using the non-normalized estimates of $\boldsymbol{B_0}$ 'passes' the scaling on to the IRF matrix through $\tilde{\boldsymbol{\Theta}}_\tau = \boldsymbol{\Phi}_\tau \tilde{\boldsymbol{B}_0}^{-1} = \boldsymbol{\Phi}_\tau \boldsymbol{B_0}^{-1} \boldsymbol{C}^{-1} = \boldsymbol{\Theta}_\tau \boldsymbol{C}^{-1}$. But this is the same for each horizon $\tau = 0, 1, ..., H$, so the whole function is just resized compared to its true value, and the shape preserved. As there is no guarantee that the obtained $\boldsymbol{C}$ will be the same for both variants of the model (IG and HS), for a better comparison of the IRFs it is better to use the normalized estimate of $\boldsymbol{B_0}$, which yields $\boldsymbol{\Theta}_\tau^{norm} = \boldsymbol{\Phi}_\tau (\boldsymbol{B_0}^{norm})^{-1} = \boldsymbol{\Phi}_\tau \boldsymbol{B_0}^{-1} \boldsymbol{Z}^{-1} = \boldsymbol{\Theta}_\tau \boldsymbol{Z}^{-1}$. As $\boldsymbol{Z}$ includes the diagonal of the true $\boldsymbol{B_0}$, it is the same for all models.

FEVD: The Forecast Error Variance Decomposition for horizon τ is computed with the help of the Mean Squared Prediction Error (MSPE) for that horizon. The Prediction Error is defined as $\boldsymbol{y}_{t+\tau} - \boldsymbol{y}_{t+\tau|t} = \sum_{i=0}^{\tau-1} \boldsymbol{\Phi}_i \boldsymbol{u}_{t+\tau-i} = \sum_{i=0}^{\tau-1} \boldsymbol{\Theta}_i \boldsymbol{\varepsilon}_{t+\tau-i}$, hence the MSPE for τ is given by

$$\begin{aligned} MSPE(\tau) &= \mathbf{E}((\boldsymbol{y}_{t+\tau} - \boldsymbol{y}_{t+\tau|t})(\boldsymbol{y}_{t+\tau} - \boldsymbol{y}_{t+\tau|t})') \\ &= \sum_{i=0}^{\tau-1} \boldsymbol{\Phi}_i \boldsymbol{\Sigma}_{t+\tau-i} \boldsymbol{\Phi}_i' = \sum_{i=0}^{\tau-1} \boldsymbol{\Phi}_i \boldsymbol{B_0}^{-1} \boldsymbol{D}_{t+\tau-i} (\boldsymbol{B_0}^{-1})' \boldsymbol{\Phi}_i' \\ &= \sum_{i=0}^{\tau-1} \boldsymbol{\Theta}_i \boldsymbol{D}_{t+\tau-i} \boldsymbol{\Theta}_i' \end{aligned}$$

Then, the MSPE of $y_{j,t}$, $MSPE^j(\tau)$, is represented by the j-th diagonal element of MSPE(τ), $MSPE^j(\tau)$, which is equal to the sum of the individual cumulative contributions of each structural shock k at time t to the MSPE of variable j at time $t + \tau$, each

given by

$$MSPE_k^j(\tau) = \sum_{i=0}^{\tau-1} \theta_{jk,i}^2 e^{h_{k,t+\tau-i}}, j, k = 1, ..., n$$

The relative contribution of shock k to variable j, for horizon τ is given by $\frac{MSPE_k^j(\tau)}{MSPE^j(\tau)}$.

Regarding the scaling/normalization of $\boldsymbol{B_0}$, using the non-normalized estimates, given the available $\tilde{\boldsymbol{D}}_t$ draws, the MSPE(τ) components are

$$\begin{aligned}\tilde{\boldsymbol{\Theta}}_i \tilde{\boldsymbol{D}}_{t+\tau-i} \tilde{\boldsymbol{\Theta}}_i' &= \boldsymbol{\Phi}_i \boldsymbol{B_0}^{-1} \boldsymbol{C}^{-1} \boldsymbol{C} \boldsymbol{D}_{t+\tau-i} \boldsymbol{C}' (\boldsymbol{C}^{-1})' (\boldsymbol{B_0}^{-1})' \boldsymbol{\Phi}_i' \\ &= \boldsymbol{\Phi}_i \boldsymbol{B_0}^{-1} \boldsymbol{D}_{t+\tau-i} (\boldsymbol{B_0}^{-1})' \boldsymbol{\Phi}_i' = \boldsymbol{\Theta}_i \boldsymbol{D}_{t+\tau-i} \boldsymbol{\Theta}_i'\end{aligned}$$

Using the normalized $\boldsymbol{B_0}$ estimates, however, computes

$$\boldsymbol{\Theta}_i^{norm} \tilde{\boldsymbol{D}}_{t+\tau-i} (\boldsymbol{\Theta}_i^{norm})' = \boldsymbol{\Phi}_i \boldsymbol{B_0}^{-1} \boldsymbol{Z}^{-1} \boldsymbol{C} \boldsymbol{D}_{t+\tau-i} \boldsymbol{C}' (\boldsymbol{Z}^{-1})' (\boldsymbol{B_0}^{-1})' \boldsymbol{\Phi}_i' \neq \boldsymbol{\Theta}_i \boldsymbol{D}_{t+\tau-i} \boldsymbol{\Theta}_i'$$

So, to obtain the correct FEVD, one must use the non-normalized estimate of $\boldsymbol{B_0}$.

HDs: The Historical Decomposition of variables $\boldsymbol{y}_t$ rests on the notion that $\boldsymbol{y}_t = \sum_{s=0}^{t-1} \boldsymbol{\Theta}_s \boldsymbol{\varepsilon}_{t-s} + \boldsymbol{y}_0$. The first term represents the cumulative contribution of structural shocks since the beginning of the sample. For each variable j, the cumulative effect of shock k at time $t = 1, ..., T$ is given by

$$\hat{y}_{jt}^{(k)} = \sum_{i=0}^{t-1} \theta_{jk,i} \varepsilon_{k,t-i}$$

A time-varying $\boldsymbol{\Sigma}_t$ is of no consequence to estimating $\hat{\boldsymbol{y}}_t$, as it does not enter the formula.

The scaling of $\boldsymbol{B_0}$ is indifferent here as well, since $\tilde{\boldsymbol{\Theta}}_s = \boldsymbol{\Theta}_s \boldsymbol{C}^{-1}$ simply represents a multiplication of each column k of $\boldsymbol{\Theta}_s$ by $\frac{1}{c_k}, k = 1, ..., n$. Since the structural shock estimates computed with $\tilde{\boldsymbol{B}}_{\boldsymbol{0}}$ become $\tilde{\boldsymbol{\varepsilon}}_t = \tilde{\boldsymbol{B}}_{\boldsymbol{0}} \boldsymbol{u}_t = \boldsymbol{C} \boldsymbol{B_0} \boldsymbol{u}_t = \boldsymbol{C} \boldsymbol{\varepsilon}_t$, which is a scaled

version of each $\varepsilon_{k,t}$ by c_k, we have

$$\hat{y}_{jt}^{(k)}|\tilde{\boldsymbol{B}}_{\mathbf{0}} = \sum_{i=0}^{t-1} \tilde{\theta_{jk,i}}\tilde{\varepsilon_{k,t-i}} = \sum_{i=0}^{t-1} \theta_{jk,i}\frac{1}{c_k} \cdot c_k\varepsilon_{k,t-i} = \hat{y}_{jt}^{(k)}|\boldsymbol{B}_{\mathbf{0}}$$

A similar equality holds for $\hat{y}_{jt}^{(k)}|\boldsymbol{B}_{\mathbf{0}}^{norm}$ through an equivalent mechanism. So, it is indifferent to use the normalized or non-normalized estimates of $\boldsymbol{B}_{\mathbf{0}}$ in the computation of Historical Decompositions for this model, in principle, as long as the same version of the impact matrix is used to compute the structural shocks.

5 Simulation Study

This section compares the usage of the flexible variance version of the VAR (with the Horseshoe prior for $\omega_{i,t}$) against the tighter variance version (with the inverse Gamma prior) for estimation on artificial data. For simplicity, a bivariate VAR is used.

5.1 Simulation Setup

The performance of both versions is assessed in a simulation study using six different Data Generation processes (DGPs) for the log-volatility which mimic possible behavior of real-world data. The study consists of a Monte Carlo simulation for each DGP. In each simulation, $R = 500$ bivariate log-volatility series with $T = 400$ observations are produced, and from each of these a series of structural errors and then a corresponding bivariate VAR(2) process $\boldsymbol{y}_t$ are generated according to Equation 1 with

$$\boldsymbol{a} = \begin{bmatrix} 2 \\ 8 \end{bmatrix}, \quad \mathbf{A}_1 = \begin{bmatrix} 0.2 & 0.05 \\ -0.1 & 0.3 \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} 0.1 & 0.2 \\ -0.15 & 0.3 \end{bmatrix}$$

The inverse impact matrix and the corresponding normalized matrix are, respectively

$$\boldsymbol{B}_{\mathbf{0}} = \begin{bmatrix} 3 & 1.5 \\ -0.5 & 2 \end{bmatrix}, \quad \boldsymbol{B}_{\mathbf{0}}^{norm} = \begin{bmatrix} 1 & 0.5 \\ -0.25 & 1 \end{bmatrix}$$

The first two points of $\boldsymbol{y}_t$ are equal to the intercepts.

The structural errors $\boldsymbol{\varepsilon}_t$ are randomly drawn from $\mathcal{N} \sim (\mathbf{0}, \boldsymbol{D}_t)$, and the log-volatility processes for each simulation are generated as follows:

DGP1: each $\{\boldsymbol{h}_i\}$ follows exactly Equation 2, with $\phi_i = 0.95, i = 1, 2$, and the variance of the log-volatility error $\xi_{i,t}$ is set as

$$\omega_{i,t}^2 = \begin{cases} 0.05, & \text{if } t \leq 0.8 \cdot T, \\ 0.2, & \text{if } t > 0.8 \cdot T. \end{cases}$$

This DGP follows exactly the theoretical behavior of the model, in that the change in structural shock behavior is controlled by an increased variance of shock variance. Note that this does not produce only scenarios like the pandemic shock, where the implicit change was towards a higher shock variance, at a certain moment. Here, datasets where the variance $e^{h_{i,t}}$ decreases are also possible.

DGP2: this process features a change in behavior at the 80% mark of the series as well, but here this is ensured to be an increase in the variance of the structural shocks, $e^{h_{i,t}}$, therefore producing scenarios closer to what was seen during the pandemic: a one-time "jump" in the log-volatility processes $h_{i,t}$, so

$$h_{i,t} = \phi_i h_{i,t-1} + c \cdot \mathbf{1}\{t = t^*\} + \xi_{i,t}, \quad \xi_{i,t} \sim \mathcal{N}(0, \omega_{i,t}^2) \tag{10}$$

with $c = 2$, $\phi_i = 0.95$ again, $t^* = 0.8 \cdot T = 320$ and $\omega_{i,t}^2$ is kept constant at 0.05 for simplicity (although implied larger at the time of the jump).

DGP3: here, an artificially large value is introduced in the $h_{i,t}$ series, again at $t^* = 0.8 \cdot T$. This is helpful to assess how the model versions perform in the case of a sudden, very large (relative to the rest) value in $h_{i,t}$, reflected in a visible disturbance in $\boldsymbol{y}_t$. So, the log-volatility follows the process

$$hi, t = \begin{cases} \phi_i h_{i,t-1} + \xi_{i,t}, & \text{if } t \neq t^*, \\ 4, & \text{if } t = t^*. \end{cases} \tag{11}$$

with ϕ_i and $\omega_{i,t}^2$ as before.

DGP4: it is also of interest to find out if the model can successfully disentangle "jumps" in $h_{i,t}$ when they occur at close but different points in time for each structural shock series. So, this DGP follows also the process described in Eq. 11, but with

$$t^* = \begin{cases} 0.8 \cdot T, & \text{if } i = 1, \\ 0.8 \cdot T + 4, & \text{if } i = 2. \end{cases}$$

DGP5: this is similar to DGP4, but the log-volatility of the second structural shock is the first to see a jump, so here

$$t^* = \begin{cases} 0.8 \cdot T + 4, & \text{if } i = 1, \\ 0.8 \cdot T, & \text{if } i = 2. \end{cases}$$

DGP6: finally, in order to assess the performance of the models in case of misspecification, a simulation is held for the case where $h_{i,t}$ follows a Random Walk process. So, the log-volatility processes are generated according to

$$h_{i,t} = h_{i,t-1} + \xi_{i,t}, \quad \xi_{i,t} \sim \mathcal{N}(0, \omega_{i,t}^2) \tag{12}$$

where $\omega_{i,t}^2 = 0.05$ again. To avoid explosiveness of $h_{i,t}$, only processes where all the $T = 400$ observations are between -6 and 6 are accepted.

To ensure convergence of the MCMC chains in all simulations, the prior variance for the $\boldsymbol{B_0}$ rows used was $v_b = 0.1$. [4]

5.2 Illustration

First, to get a better idea of how each model captures the change in behavior of the processes, representative $\boldsymbol{h}^{(r)}$ of each DGP and the respective estimates (median of draws) and credible regions obtained with each model are examined.

[4]For larger values of v_b, parameter estimates for DGP1 process in particular struggled to converge.

Both models follow the true process well. The HS model seems to produce overall smoother estimated $\boldsymbol{h}$ processes than the IG, but with sharper jumps when the true h_{it} changes suddenly. Particularly, the HS model is better able to follow the sharp jumps in the true log-volatility. Not only the estimated h_{it} are closer to the true value at jump times, but the increase is (usually) at that time exactly. The IG model in contrast, already shows higher values in the periods before the jump. Figure 4 shows the $\boldsymbol{h}$ processes of the 250th generated dataset with DGP4 and respective estimated $h_i, i = 1, 2$ with each model. This discrepancy between models is clear.

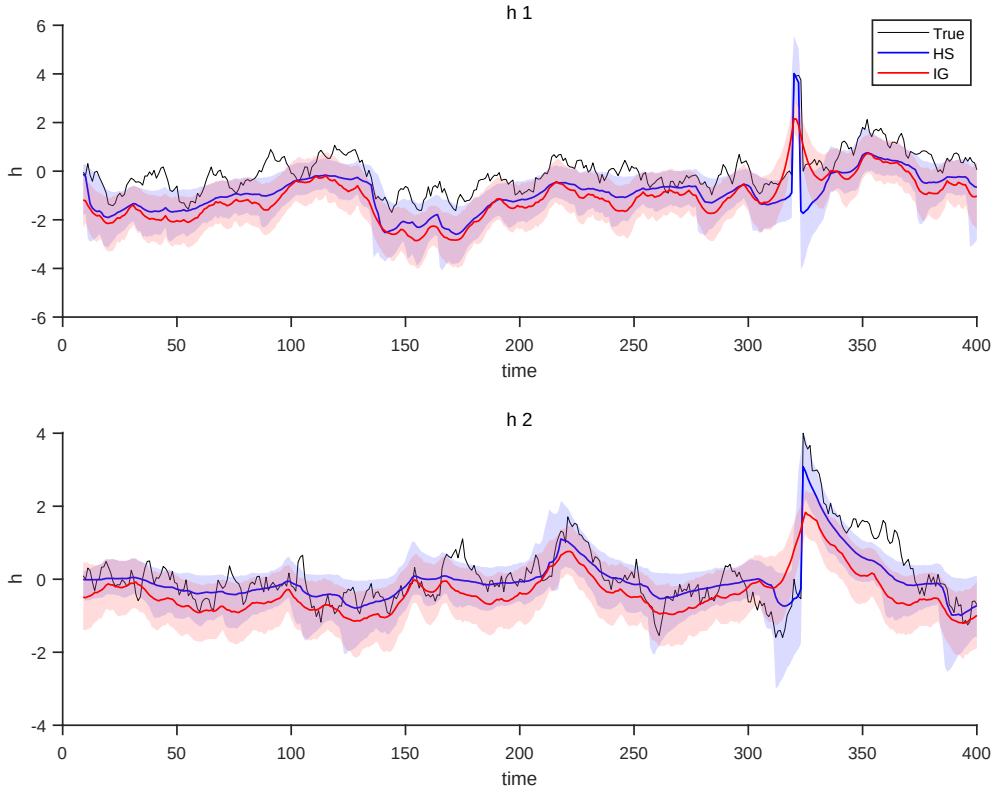


Figure 4: 250th generated log-volatility process from DGP4 and respective estimates with HS and IG model. The full lines represent the median of draws, and the shaded areas the 90% credible regions for each estimate. For both $\boldsymbol{h}_i$ it is clear the IG model estimates a slight rise just before the jump, which is not present with the HS model.

The HS model is not always capable of capturing a sizable jump in $\boldsymbol{h}$ either. Figure 5 shows an example log-volatility process from DGP5 where the HS model does show a sharp jump at the correct time, but is not capable of capturing the correct size. The

estimate is still closer than with the IG model, though. Figure 6 shows another example, this time from DGP2, where either model fails to capture the jump at all. The true $h_{2,t}$ has a sharp jump at $t = 320$, but both estimates fail to capture it.

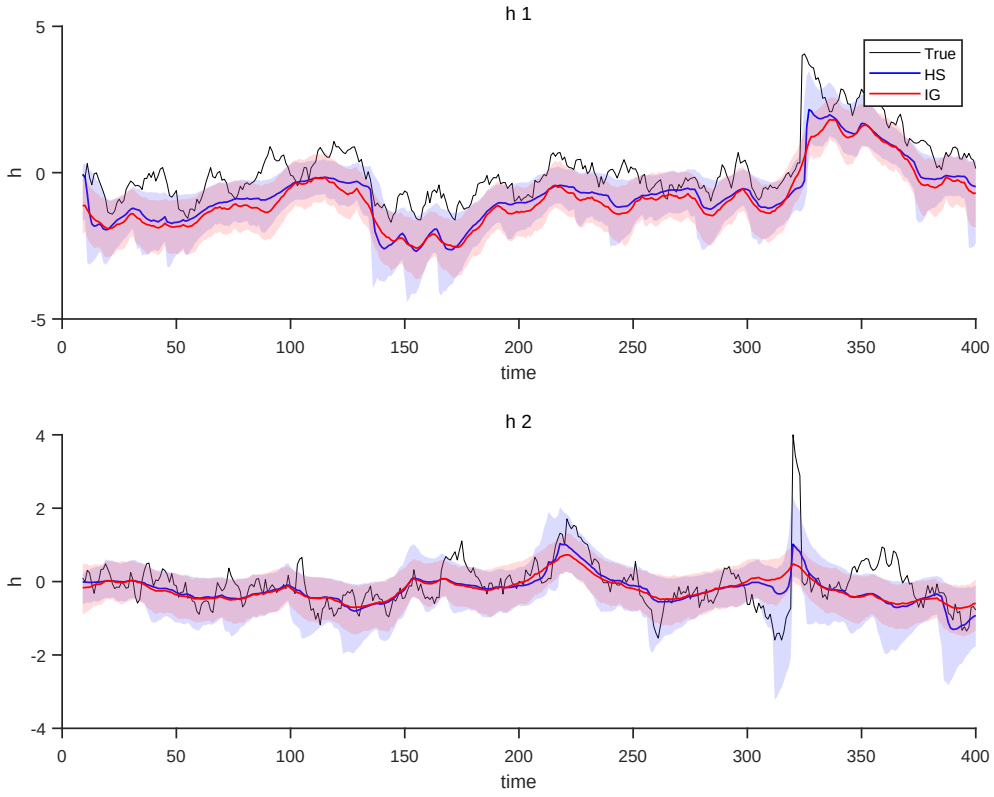


Figure 5: 250th generated log-volatility process from DGP5 and respective estimates with HS and IG model. The full lines represent the median of draws, and the shaded areas the 90% credible regions for each estimate. Neither model fully captures the jump size.

5.3 Simulation Results

In each simulation, for each of the $R = 500$ simulated datasets $\boldsymbol{y} = (\boldsymbol{y}_1, ..., \boldsymbol{y}_T)$ the VAR coefficients, the normalized (inverse) impact matrix, and the log-volatility processes are estimated with both the HS and the IG model. The resulting estimates are evaluated, for each model, for each DGP, with the respective mean absolute error and mean squared

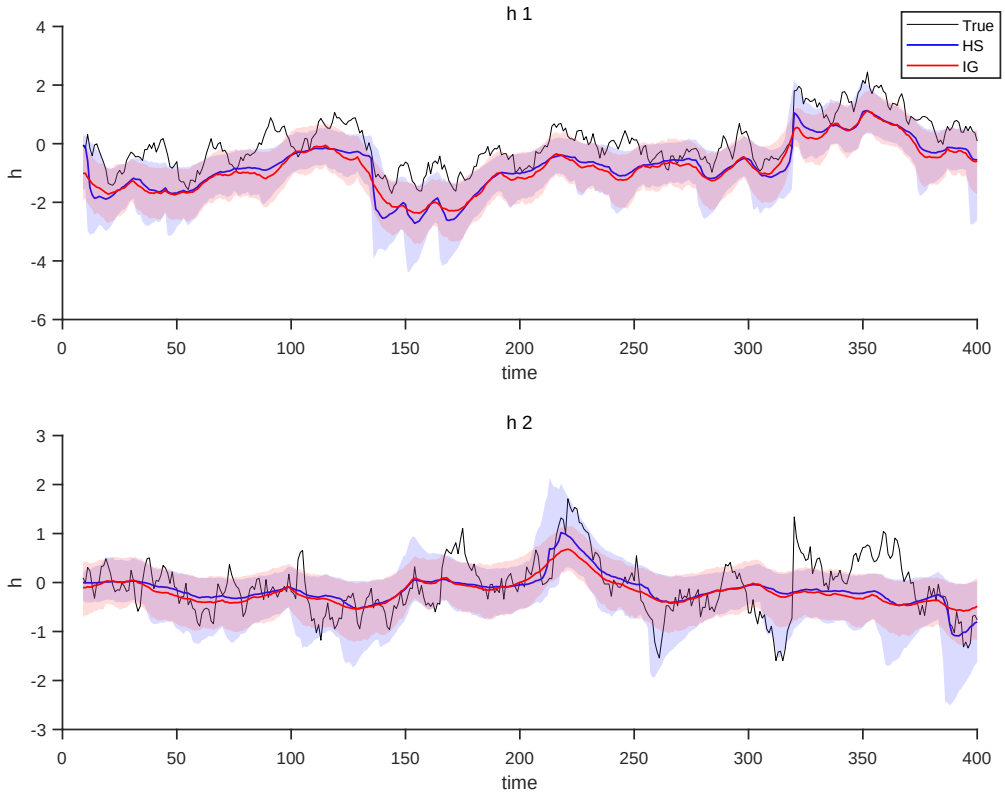


Figure 6: 250th generated log-volatility process from DGP2 and respective estimates with HS and IG model. The full lines represent the median of draws, and the shaded areas the 90% credible regions for each estimate. Both models fail to capture the jump.

error metrics, defined as

$$MAE_{\alpha_i} = \frac{1}{R}\sum_{r=1}^{R}|\hat{\alpha_i}^{(r)} - \alpha_i|\,, \quad MSE_{\alpha_i} = \frac{1}{R}\sum_{r=1}^{R}(\hat{\alpha_i}^{(r)} - \alpha_i)^2$$

$$MAE_{b_{ij}} = \frac{1}{R}\sum_{r=1}^{R}|\hat{b_{ij}}^{(r)} - b_{ij}|\,, \quad MSE_{b_{ij}} = \frac{1}{R}\sum_{r=1}^{R}(\hat{b_{ij}}^{(r)} - b_{ij})^2$$

$$MAE_{\sigma_{ij,t}} = \frac{1}{R}\sum_{r=1}^{R}|\hat{\sigma_{ij,t}}^{(r)} - \sigma_{ij,t}|\,, \quad MSE_{\sigma_{ij,t}} = \frac{1}{R}\sum_{r=1}^{R}(\hat{\sigma_{ij,t}}^{(r)} - \sigma_{ij,t})^2$$

where α_i is the i-th of the $k = n(n \cdot p + 1)$ entries of $\boldsymbol{\alpha}$, so the intercepts $\boldsymbol{a}$ and the autoregressive coefficients (entries of $\boldsymbol{A}_j, j = 1, ..., p$), b_{ij} is the $[ij]$ entry of $\boldsymbol{B_0}^{norm}$, $\sigma_{ij,t}$ is the $[ij]$ entry of $\boldsymbol{\Sigma}_t = \boldsymbol{B_0}^{-1}\boldsymbol{D}_t(\boldsymbol{B_0}^{-1})'$, and $\hat{\alpha_i}^{(r)}$, $\hat{b_{ij}}^{(r)}$ and $\hat{\sigma_{ij,t}}^{(r)}$ their respective estimates

(median of draws) for the r-th generated dataset. The entries of $\boldsymbol{\Sigma}_t$ are examined instead of the $\boldsymbol{h}_t$ values because they are not subject to incorrect scaling, which could confuse the error metrics.

Further, average values are computed for each of those quantities. The values for $MAE_{b_{ii}}, MSE_{b_{ii}}, i = 1, ..., n$ are not included in this average, because we already know them to be very near zero.

$$MAE_\alpha = \frac{1}{k}\sum_{i=1}^{k} MAE_{\alpha_i}, \quad MAE_b = \frac{1}{n(n-1)}\sum_{i=1}^{n}\sum_{j=1,j\neq i}^{n} MAE_{b_{ij}}, \quad MAE_{\sigma_{ij}} = \frac{1}{T}\sum_{t=1}^{T} MAE_{\sigma_{ij,t}}$$

$$MSE_\alpha = \frac{1}{k}\sum_{i=1}^{k} MSE_{\alpha_i}, \quad MSE_b = \frac{1}{n(n-1)}\sum_{i=1}^{n}\sum_{j=1,j\neq i}^{n} MSE_{b_{ij}}, \quad MSE_{\sigma_{ij}} = \frac{1}{T}\sum_{t=1}^{T} MSE_{\sigma_{ij,t}}$$

HS

	$\boldsymbol{\alpha}$	$\boldsymbol{B_0}$	$\boldsymbol{\sigma}_{11}$	$\boldsymbol{\sigma}_{12} = \boldsymbol{\sigma}_{21}$	$\boldsymbol{\sigma}_{22}$
DGP1	0.191	0.054	0.069	0.063	0.117
DGP2	0.200	0.073	0.062	0.055	0.108
DGP3	0.199	0.076	0.097	0.087	0.174
DGP4	0.189	0.048	0.092	0.096	0.175
DGP5	0.180	0.049	0.098	0.083	0.145
DGP6	0.181	0.043	0.314	0.284	0.498

IG

	$\boldsymbol{\alpha}$	$\boldsymbol{B_0}$	$\boldsymbol{\sigma}_{11}$	$\boldsymbol{\sigma}_{12} = \boldsymbol{\sigma}_{21}$	$\boldsymbol{\sigma}_{22}$
DGP1	0.191	0.055	0.066	0.061	0.113
DGP2	0.200	0.075	0.061	0.055	0.107
DGP3	0.199	0.077	0.122	0.094	0.202
DGP4	0.192	0.053	0.099	0.110	0.197
DGP5	0.185	0.053	0.106	0.097	0.165
DGP6	0.180	0.041	0.283	0.253	0.442

Table 1: Average MAE values for $\boldsymbol{B_0}$, $\boldsymbol{\alpha}$, and entries of $\boldsymbol{\Sigma}_t = \boldsymbol{B_0}^{-1}\boldsymbol{D}_t(\boldsymbol{B_0}^{-1})'$, for each DGP. The estimate used to compute the MAE is the median of draws of each entry. The first panel contains the values computed with the HS model, the second with the IG model.

Table 1 presents the average MAE values of the HS and IG models for all DGPs. A breakdown of individual MAE estimates and all MSE metrics can be found in Appendix

D.

The HS model achieves slightly smaller average MAE values than the IG model in the estimates of the normalized $\boldsymbol{B_0}$ entries for all but the last DGP. The differences, however, are on a very small scale. The estimates of the $\boldsymbol{\alpha}$ entries also show very similar average MAE for all DGPs, with a slight advantage of the HS model for DGPs 4 and 5, and again better performance of the IG model with DGP6. The differences are, again, of a very small order.

Thus, regarding VAR parameters and impact matrix estimates, there seems to be no very large differences in performance between both models, for the 6 analysed DGPs.

Examining now the average MAE for the entries of $\boldsymbol{\Sigma}_t = \boldsymbol{B_0}^{-1}\boldsymbol{D}_t(\boldsymbol{B_0}^{-1})'$, the IG model seems to perform slightly better with DGPs 1, 2 and 6. With the other DGPs, in particular with DGP 3, the HS model performs better. Another interesting point is the difference in average MAE between the variances of each variable: it is larger for the σ_{22} parameter than for σ_{11}, indicating that both models do systematically worse when it comes to estimating the variance of $y_{2,t}$, regardless of the DGP.

The reported MAE values regarding the entries of $\boldsymbol{\Sigma}_t$ represent an average over the whole sample. This is a somewhat large period, so the average might not be representative of the performance of the models particularly at the points of most interest, the "sudden jumps" in variance. These take place, for all DGPs, at the $t^* = 320$ mark (and $t^* = 324$ for DGPs 4 and 5). It is therefore of interest to examine the $MAE_{\sigma_{ij,t}}$ values around $T = 320$, to examine how well each model captures the change in behavior of the processes. Figure 7 shows the individual MAE for $\sigma_{11,t}$ and $\sigma_{22,t}$ of each model for DGP3. The values are noticeably larger with the IG model at the time of the jump, which indicates the HS model is better at approximating the variance when such an event takes place. This is in line with the exemplary $\boldsymbol{h}$ processes examined previously.

Moreover, there is again a small but noticeable increase of the MAE for the IG model already before the jump, which is not visible for the HS model. This is also the case for DGPs 4 and 5 (see Appendix), which show much higher individual MAE values at moment of the jumps, and an increase of MAE of IG model $\sigma_{ij,t}$ estimates just before.

The case is slightly different when the jump is smaller: with DGP2 we can see also an increase in error size before the jump, but similar in both models. At the jump,

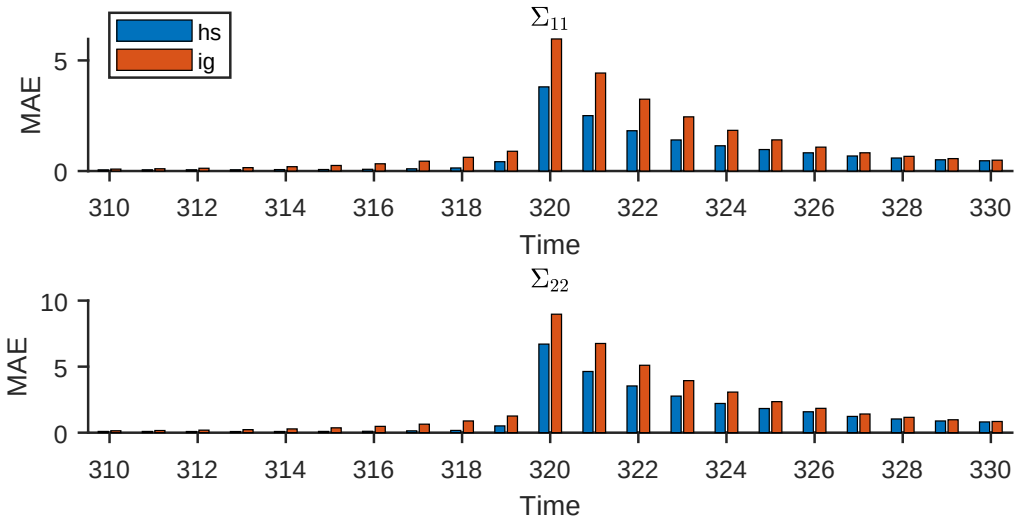


Figure 7: Individual MAE values for the $\boldsymbol{\Sigma}_t = \boldsymbol{B_0}^{-1}\boldsymbol{D}_t(\boldsymbol{B_0}^{-1})'$ diagonal entries, for 10 periods before and after $T = 320$, for DGP3. The estimates taken for the MAE computation are the medians of draws the respective entry of $\boldsymbol{\Sigma}_t$. The IG model shows higher values at the time of the jump.

however, the MAE is again noticeable larger for the IG model. Figure 8 shows the MAE quantities for each model around $T = 320$ for the simulation with DGP2. It could be the case that, like in the example shown in Fig. 6, both models fail to capture the jump in several datasets from this DGP, creating a large error before, during and after that moment.

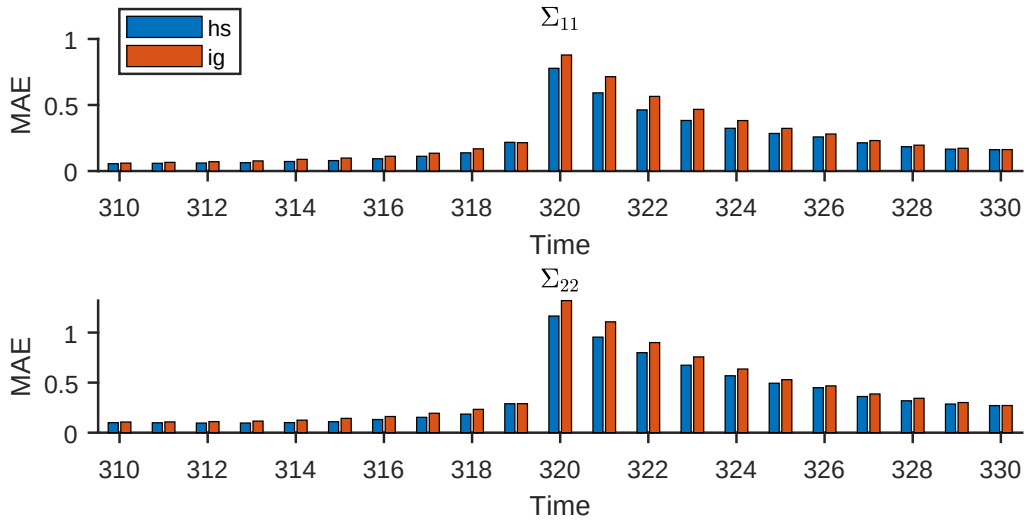


Figure 8: Individual MAE values for the $\boldsymbol{\Sigma}_t = \boldsymbol{B_0}^{-1}\boldsymbol{D}_t(\boldsymbol{B_0}^{-1})'$ diagonal entries, for 10 periods before and after $T = 320$, for DGP2. The estimates taken for the MAE computation are the medians of draws the respective entry of $\boldsymbol{\Sigma}_t$. Both models show a similar increase in error before the jump, but higher for IG after.

6 Empirical Analysis

This section describes a real-world application of the methodology developed in the previous pages.

Both the HS and the IG model are applied to the problem of disentangling oil market variables' movements into supply and demand components. The shocks are labeled with the help of external instruments and a narrative measure.

The oil market, particularly the question of price decomposition, has been widely assessed in the literature (Kilian (2008), Baumeister and Hamilton (2019) ,Antolín-Díaz and Rubio-Ramírez (2018)), with many different estimations of the relative role of demand and supply shocks on the global price of oil.

Following Braun and Brüggemann (2022), the global oil market is modeled with a four equation VAR(13), where $y_t = (prod_t, rea_t, rpo_t, i_t)'$, with $prod_t$ representing the log of world oil production, rea_t the log of industrial production index from Baumeister and Hamilton (2019), rpo_t the log of the real price of oil and, finally, i_t representing the seasonally adjusted log of OECD crude oil inventories.

6.1 Dataset

The Braun and Brüggemann (2022) sample (from now on designated BB22 sample) includes monthly data from 1978M08 to 2018M11. The main sample used here recreates the BB22 sample for the period 1988M01 to 2025M05. Before 1988, data for the correct computation of the i_t series is not available. Each series in y_t is computed according to the descriptions given in Braun and Brüggemann (2022) and Baumeister and Hamilton (2019), using publicly available data.

The Oil Production data is taken directly from the U.S. Energy Information Administrations (EIA) database. The Real Price of Oil is obtained by deflating the West Texas Intermediate (WTI) nominal spot price series (also from EIA) with the US Consumer Price Index (CPIAUCSL) taken from the Federal Reserve Economic Data (FRED) database. The Industrial Production index is taken directly from Christiane Baumeister's website. The OECD crude Oil Inventories must be approximated. This is done as described in Baumeister and Hamilton (2019), Appendix C. The US crude oil inventories are multiplied by the ratio of *OECD inventories of crude petroleum and petroleum products* series to *U.S. inventories of petroleum and petroleum products*. The series is then seasonally adjusted with the X-13ARIMA-SEATS method. All these variables are obtained from the EIA database as well. The sources used only allow a plausible sample from 1988M01 onward, so this is where the sample starts.

6.2 Structural Shocks and Tools

Before diving into the results, it is appropriate to discuss how shocks are obtained given all model draws.

The estimates of structural shocks are obtained by computing the shock series for each draw d of model parameters, $\hat{\boldsymbol{\varepsilon}}_t^{(d)}$, and then taking the mean over all draws. The series for each draw is obtained as

$$\begin{aligned}\hat{\boldsymbol{\varepsilon}}_t^{(d)} = \hat{\boldsymbol{B}}_{\mathbf{0}}{}^{(d)} \cdot \hat{\boldsymbol{u}}_t^{(d)} &= \hat{\boldsymbol{B}}_{\mathbf{0}}{}^{(d)} \cdot (\boldsymbol{y}_t - \hat{\boldsymbol{y}}_t^{(d)}) \\ &= \hat{\boldsymbol{B}}_{\mathbf{0}}{}^{(d)} \cdot (\boldsymbol{y}_t - \hat{\boldsymbol{a}}_t^{(d)} - \hat{\boldsymbol{A}}_1^{(d)}\boldsymbol{y}_{t-1} - \hat{\boldsymbol{A}}_2^{(d)}\boldsymbol{y}_{t-2} - ... - \hat{\boldsymbol{A}}_p^{(d)}\boldsymbol{y}_{t-p})\end{aligned}$$

where $\hat{\boldsymbol{B}}_{\boldsymbol{0}}^{(d)}$ is the draw d of the inverse impact matrix, $\hat{\boldsymbol{a}}_t^{(d)}$ is the vector of intercepts drawn and $\hat{\boldsymbol{A}}_i^{(d)}, i = 1, ..., p$ are the matrices of drawn autoregressive coefficients corresponding to each lag i.

The version of the impact matrix used here is the non-normalized one, so corresponding to $\tilde{\boldsymbol{B}}_{\boldsymbol{0}}$. This means that the shocks computed are scaled in relation to the real shocks by unknown constants $c_i, i = 1, ..., n$. For ease, here they are represented simply as $\hat{\boldsymbol{\varepsilon}}_t$.

The estimated structural shocks are shown in Figure 9.

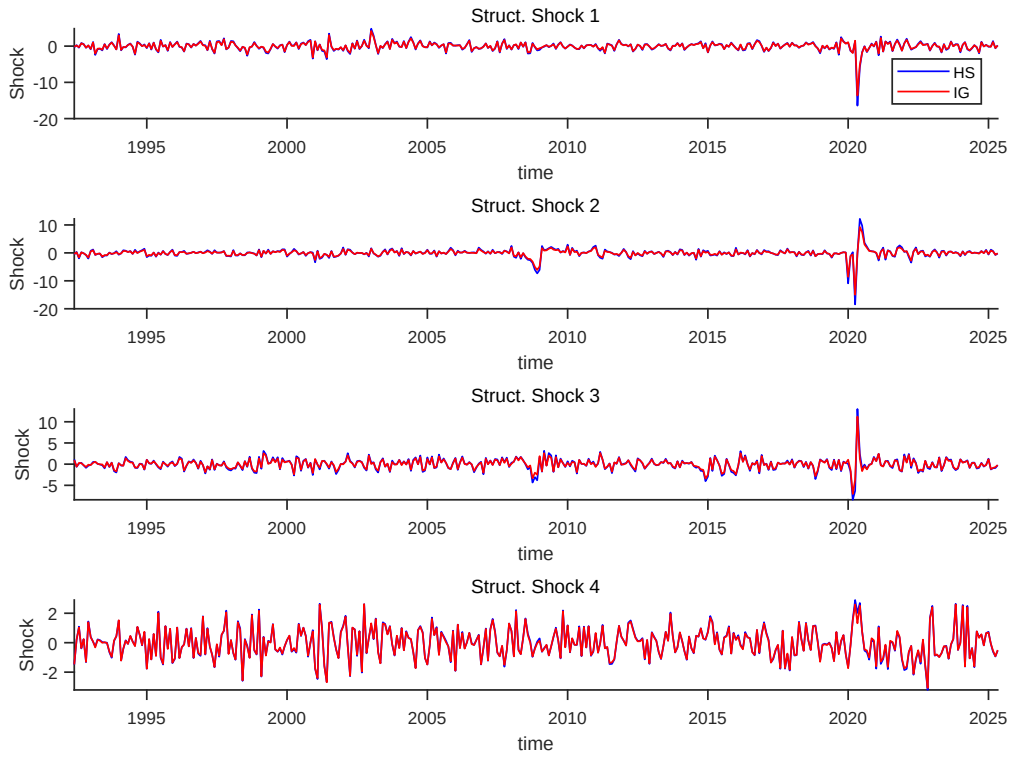


Figure 9: Estimated structural shocks with the HS and IG models. Each shock series is the median of computed shocks across all model draws.

The Forecast Error Variance Decompositions, like the Impulse Responses, are computed draw by draw. The decompositions are obtained from the draws of $\boldsymbol{B}_{\boldsymbol{0}}$ and VAR coefficients, and an estimated average path $\hat{\boldsymbol{h}}_T, \hat{\boldsymbol{h}}_{T+1}, ..., \hat{\boldsymbol{h}}_{T+\tau}$, for horizon τ, which enters the FEVD through $\hat{\boldsymbol{D}}_t = diag(e^{\hat{h}_{1t}}, ..., e^{\hat{h}_{nt}})$. This average path is computed from the

draw $\boldsymbol{h}_T^{(d)}$ recursively as $h_{i,t+1}^{(d)} = \phi_i^{(d)} h_{i,t}^{(d)} + \xi_{it}^{(d)}, \xi_{it}^{(d)} \sim \mathcal{N}(0, (\omega_{i,T}^2)^{(d)}), \forall t = T, ..., \tau - 1$. The implicit assumption is that ω_{it}^2 does not change and the forecast is made at the end of the sample. This is a necessary simplification, as there is no assumed path for $\omega_{i,t}$, because the local variance components are assumed independent. This means that the estimated FEVD (median across draws) is not a representation of the relative contribution of shocks to variances of each variable at all points in the past, but of how we would expect this contributions to evolve if all remained as last seen (at time T). Note also that using the median of FEVD across model draws as the estimate might mean it loses the 'sum to 1' property of the FEVDs.

The Historical Decompositions are also computed draw by draw, and the estimate of the contribution to the behavior of variable j from shock k, $j, k = 1, ..., n$ at a certain time point is the median of this contribution over all draws. This means that the estimated contributions of all shocks do not necessarily sum to the actual (demeaned and detrended) variable change relative to the beginning of the sample.

6.3 Labeling the shocks

Braun and Brüggemann (2022), following Kilian and Murphy (2014), identify three out of four structural shocks: an Oil Supply shock, a Oil Flow Demand and an Oil Inventory (or Speculative) Demand shock, while the fourth is left to capture only the residual variation in the endogenous variables. Baumeister and Hamilton (2019), making use of a similar model, identify also a Supply, Consumption Demand and Speculative Demand shock, but also a series of structural Economic Activity shocks, driven by changes to Aggregate Demand and productivity shocks.

The current analysis strives to identify the same shocks as in Baumeister and Hamilton (2019), making use of a mix of instrumental variables, following the strategy outlined in Bertsche and Braun (2020), and narrative checks.

The method consists in, first, examining the pairwise correlations between instruments z_t and each estimated structural shock $\tilde{\varepsilon_{it}}$. In theory, a shock series with high correlation with the instrument is a candidate for the label that instrument is meant to represent.

Instrument selection is notoriously tricky. According to Bertsche and Braun (2020), who follow the assumptions of Mertens and Ravn (2013), a good instrument must be both relevant, meaning it is correlated to the structural shock it should identify, and exogenous, meaning it is not correlated with the other shocks. The need for the first characteristic is obvious, as the correlation is what is used for shock labeling. The second is naturally required when the correlations are the only or major source of identifying information. However, an instrument that is correlated with more than one shock can still provide information as to the true labels of the identified shocks.

Given that this combination of variables have been used to model the global oil market in several different papers already, the conclusions from those papers are useful for labeling the obtained shocks. In fact, estimated structural shocks from previous papers can and have been used as instruments for identifying shocks in other models (Bertsche and Braun (2020)). Here, Oil Supply, Oil (Consumption and Inventory) Demand and an Economic Activity shock series from Baumeister and Hamilton (2019) [5] and all shock series obtained from replicating the analysis from Braun and Brüggemann (2022) are used.

Graphical observation (Fig. 9) already provides some information. The first three shock series all show a large one-time movement in early 2020, while the fourth shock does not. This can hardly be the supply or flow demand shock, since there is no visible effect of the COVID-19 pandemic. The worldwide slump in economic activity during the many lock-downs in the first half of 2020 caused unprecedented shocks to both supply and demand for crude oil globally.

The direction of the shocks alone does not provide a lot of information on their own, as the scaling issue could have caused the sign to be switched. However this does not affect the absolute correlation between the shock series and the instruments.

The correlation between the estimated shock series and the Baumeister shocks are reported in Table 2. The Baumeister Supply shock series is highly correlated with two of the estimated shocks, the first and third. The signs of these correlations is different but, as mentioned above, on their own the signs are not meaningful. The opposite signs can indicate that one of these shocks is the Supply shock and the other is one that tends

[5] available in Christiane Baumeister's website

to take an opposite direction to it. Examining the other correlations, it is plain to see that the Economic Activity shock has a very large correlation with the second shock. It is, therefore, sensible to take this to be the the Economic Shock in the present model. The Inventory Demand shock is uncorrelated with all but the last shock, with which it shares a nearly 0.80 correlation. So, the forth shock is a candidate for the Speculative Demand label. Lastly, the Demand shock is strongly correlated with the third estimated shock, although it also shows a non-negligible correlation with the forth and, as stated above, the third one is correlated with the Supply shock. However, the first shock, also correlated with the Supply series, has no strong correlation to any other of the Baumeister shocks. Hence, the first shock can be taken as the Supply one, and the third as the Demand.

HS

Shocks	1	2	3	4
Oil Supply	0.616	-0.158	-0.715	0.040
Consump. Demand	0.106	0.096	0.738	-0.367
Invent. Demand	0.082	-0.050	0.030	0.792
Econ. Activity	-0.181	0.906	0.181	0.045

IG

Shocks	1	2	3	4
Oil Supply	0.589	-0.182	-0.713	0.068
Consump. Demand	0.144	0.148	0.728	-0.371
Invent. Demand	0.088	-0.048	0.039	0.794
Econ. Activity	-0.212	0.902	0.069	0.055

Table 2: Correlation coefficients between the shock series from Baumeister and Hamilton (2019) and the estimated structural shocks, with the HS model (first panel) and the IG model (second panel).

The estimated shocks are, then, labeled as follows: the first is the Oil Supply shock, the second the Economic Activity one, the third is the Consumption Demand and the forth is the Inventory Demand.

It is worthwhile looking also at the shock series obtained with the Braun and Brüggemann (2022) (BB22) model for corroboration. Table 3 reports the pairwise correlations between the estimated shocks and the BB22 ones. Braun and Brüggemann (2022) use

sign restrictions and an instrument for the supply shock (the OPEC production shortfall series of Kilian (2008)) for identification, and estimate the model with six different restriction combinations. The estimation can be done with and without exogeneity and relevance constraints on the instrumental variable, and with one of three possible supply elasticity restrictions: none, one inspired by Herrera and Rangaraju (2020), called HR20, and one inspired by Baumeister and Hamilton (2019), called BH19. Here, only the versions without the instrumental variable restrictions and with the HR20 and BH19 restrictions on elasticity are tested, the reason being that the instrumental variable from Braun and Brüggemann (2022) is not available for more recent years, so using it for information would distort the estimates. The original paper finds that the HR20 and BH19 restrictions are a good addition to the model, so these are kept.

The correlations with the BB22 shocks are much weaker than the ones with the Baumeister shocks. The first shock has a similar correlation coefficient size with the Flow and the Inventory demand for this set. The second shock is most strongly correlated with the Residual Shock. Both the first and the second shocks have a similar absolute correlation to the Supply shock. The third shock is also most strongly correlated with the Inventory Demand shock, and the forth shock is uncorrelated with all BB shock series.

The correlation between estimated shocks and the Baumeister and BB shock series are presented in Table 3. Only HS shocks are reported (the IG shocks yield similar results).

To discard the possibility that the BB model does not adapt well to the pandemic period, and these observations are distorting the correlations, the model is estimated again, this time with the original Braun and Brüggemann (2022) sample, and the BB shocks are re-estimated for this period. The conclusions are similar. The pairwise correlations between the estimated and Baumeister shocks which were strong in the main sample are also strong in this older one, and the BB shocks share very weak correlations with the estimated shocks.

HR20

Shock	1	2	3	4
Supply	0.152	-0.168	-0.108	-0.009
Flow Demand	-0.192	0.118	0.088	0.014
Inventory Demand	-0.219	0.055	0.131	0.018
Residual	0.101	-0.181	-0.052	-0.007

BH19

Shock	1	2	3	4
Supply	0.123	-0.181	-0.086	-0.006
Flow Demand	-0.216	0.087	0.108	0.017
Inventory Demand	-0.218	0.050	0.144	0.019
Residual	0.142	-0.162	-0.067	-0.011

Table 3: Correlation coefficients between the shock series from Braun and Brüggemann (2022), without the IV restriction and with the HR20 (first panel) and BH19 (second panel) elasticity restrictions, and the estimated structural shocks with the HS model.

6.4 Results and Discussion

Both models are estimated with $v_b = 0.1$. The estimated (median of draws) log-volatility processes, $\hat{\boldsymbol{h}}_t$ are displayed in Figure 10.

As shown in Figure 9, the estimated structural shocks do not differ a lot between models. The extreme shocks at the start of the pandemic are slightly larger in magnitude under the HS-model, but not by much. The distribution of normalized draws of the inverse impact matrix $\boldsymbol{B}_{\mathbf{0}}^{norm}$ for each model is displayed in Figure 11. The values are close, as expected, given the nearly equivalent performance of both models in the simulations.

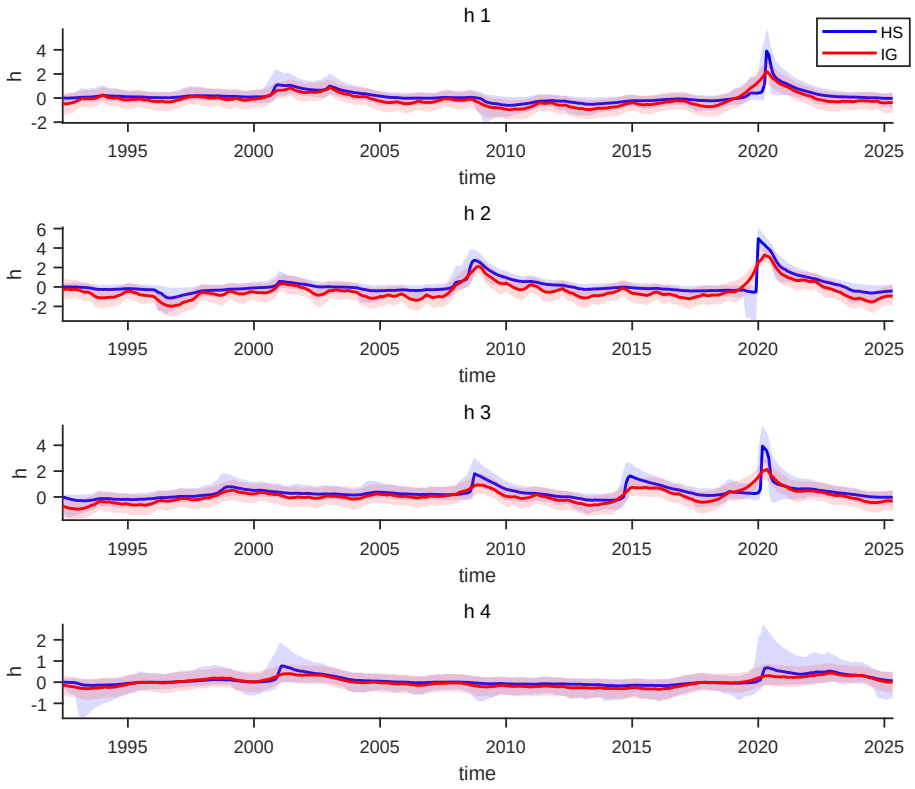


Figure 10: Log-volatility processes estimated with HS and IG models. The full lines represent the median h_{it} across all draws, and the shaded regions the 90% credible regions.

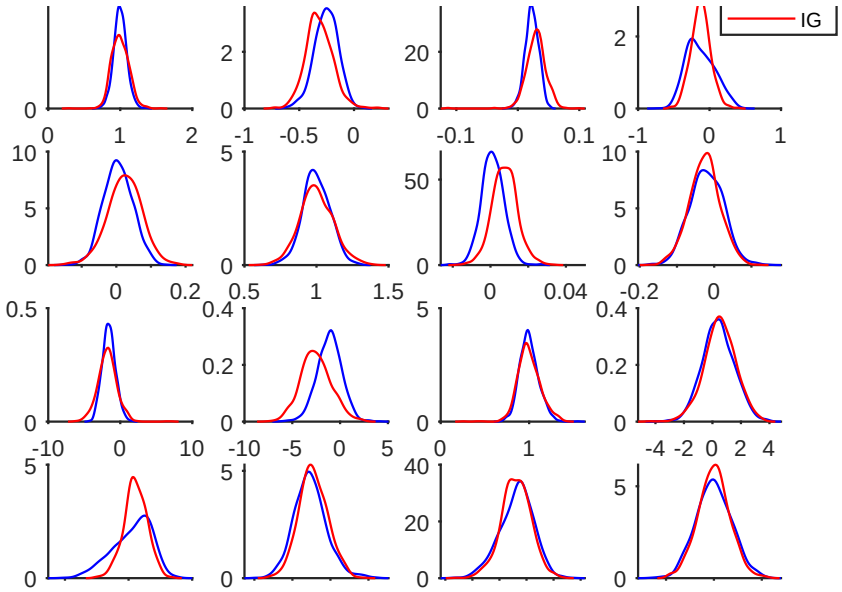


Figure 11: Densities of normalized draws of B0 entries when estimated with HS and IG models. The estimates largely coincide.

The similar $\boldsymbol{B_0}$ and structural shocks estimates are reflected in the estimated IRFs. These are reported in Figure 12. Both models yield IRFs with the same shape, with almost overlapping median responses and credible bands in a lot of cases, although the HS model's IRFs include zero in the 90% credible band more often than the ones from the IG model.

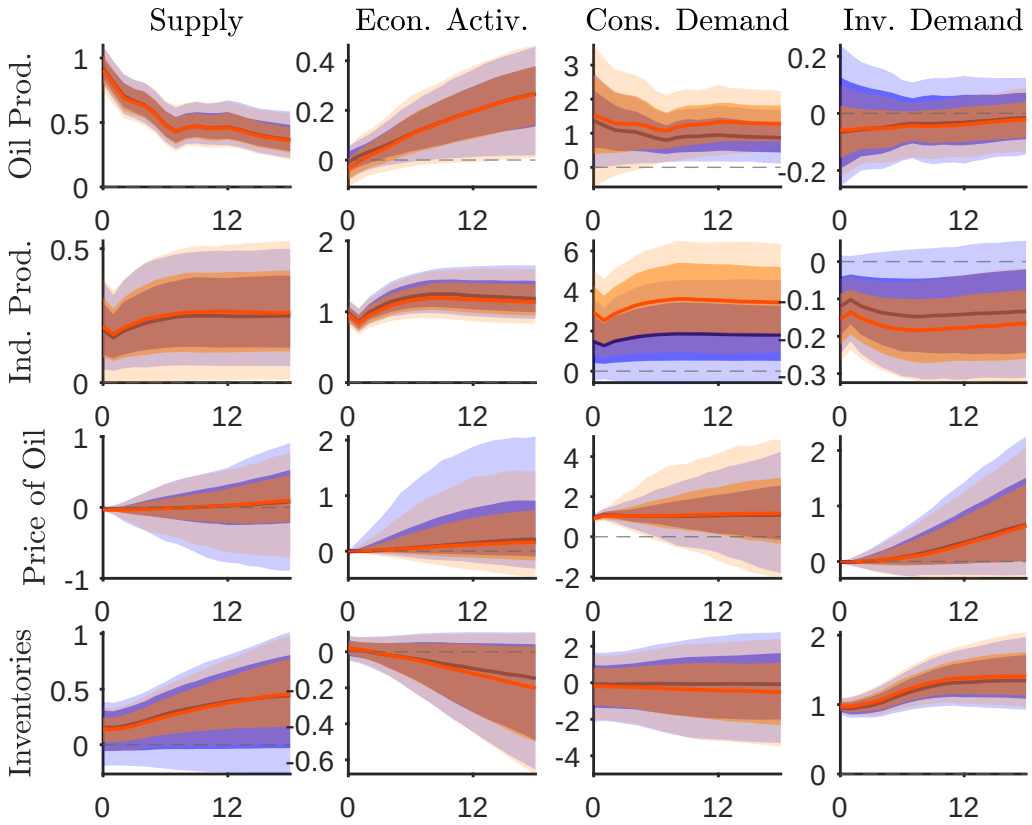


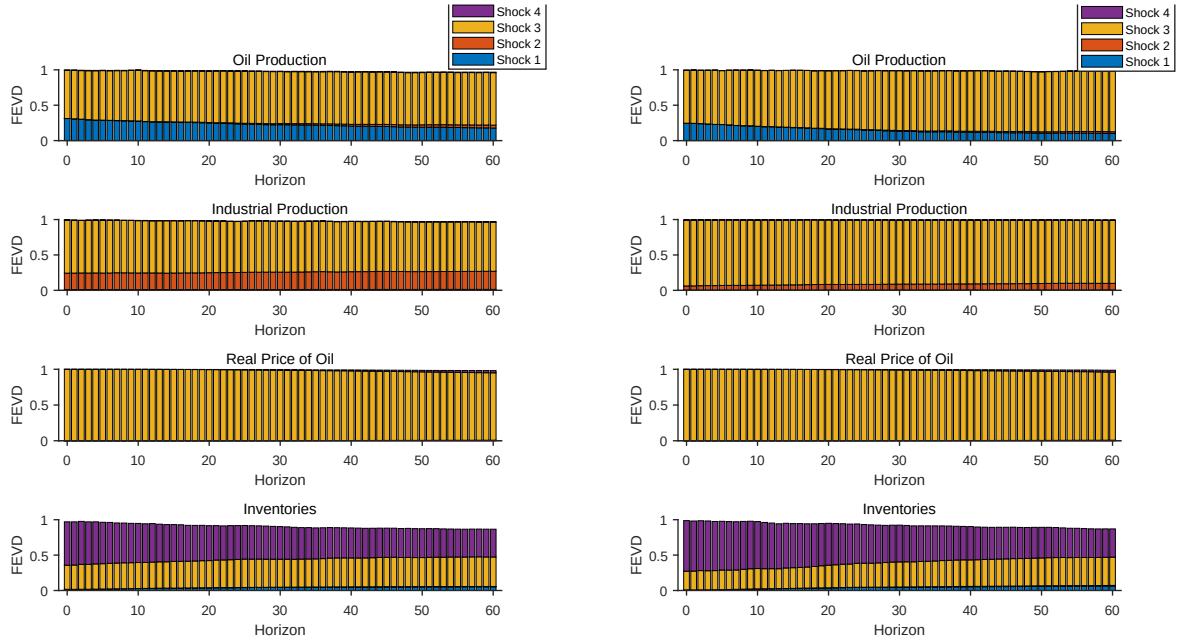
Figure 12: Impulse response functions, for the HS and IG models, until 18-month horizon. The full lines represent the median across all draws, the darker shaded regions the 68% credible regions and the lighter shaded regions the 90% credible regions. Estimates are similar between models, although the HS model is a slightly more conservative regarding effect significance.

The computed responses to the Consumption Demand shock differ the most. The IG model estimates a higher response on impact for the real activity variable, and the 90% credible bands do not include zero for any of the computed horizons. The same IRF, when computed with the HS model draws, does not rule out that the response is zero for all horizons. A similar situation is observed for the responses of Industrial Production to the Inventory Demand shock, of Oil Production to Consumption Demand shocks, and of

Oil Stocks to Supply shocks. For all other cases, even when the median response differs between models, the credible bands mostly overlap, and there are not other instances of one model including zero in the credible band while the other does not.

As an additional check on the shock labels, the impulse responses of shock 1 match in shape those estimated by Juvenal and Petrella (2015) for the Supply shock on Oil production, inventories, oil prices and industrial production, with the sign switched. Also, the shapes of all estimated IRFs match those obtained by Baumeister and Hamilton (2019), with switched signs for the Supply and Inventory Demand shocks. This is further evidence towards the labels obtained in Section 6.3.

The estimated forecast error variance decompositions are shown in Figure 13. The FEVD for the first two variables (World Oil productions and Industrial Production) in the VAR are rather different between models.



(a) HS model

(b) IG model

Figure 13: Forecast error variance decompositions, for the HS and IG models, until 60-month horizon. Each estimated relative variance contribution is computed as the median contribution across all draws. The "sum to 1" property does not necessarily hold. Shock labels: 1 - Oil Supply; 2 - Economic Activity; 3 - Consumption Demand; 4 - Inventory Demand.

The IG model estimates a much larger relative importance of the Oil Consumption Demand shocks for the variance of both of these variables. The HS model, while still

estimating the Consumption Demand to have the largest importance, attributes also a large part of variance in the first variable to Oil Supply shocks, and even some to Economic Activity shocks, in later horizons. For the Industrial Production variable, the Economic Activity shocks are found to have higher relative importance than with the IG model. This decomposition makes perhaps a bit more economic sense, due to the relationship between Output and aggregate Demand or productivity, and the Economic Activity shock series captures precisely those variables. Both models estimate the variance in Oil Prices to be driven almost exclusively by Oil Demand Shocks. The decomposition for the Stocks variable is also rather similar between models.

The computed Historical Decompositions, reported in Figures 14 and 15, are also different between models.

Regarding the first variable, World Oil production, the IG model attributes a larger contribution to the deviations in this variable to the third structural shock, Oil Demand, during the 2008-2014 period, in comparison with the HS model. Around the start of the pandemic, both models identify first a large negative impact of the Demand shock, and of Supply right after. The IG model estimates larger negative contributions of the Economic Activity shocks from 2008 until the end of the sample. Regarding the Industrial Production variable, both models find a small negative contribution of Oil Supply shocks after the early pandemic period. The Oil Demand shocks are again found to have a larger positive contribution with the IG model than with the HS one. For the third variable, measuring the Real Price of Oil, the differences between estimated contributions with each model are mostly of magnitude. Both find similar contributions of shocks for the same periods, notably a large negative contribution of the Oil Demand at the start of the pandemic. Both also identify a weakening of Oil Demand around mid-2014. This is consistent with the findings from Baumeister and Kilian (2015), which attribute the oil price collapse of 2014-2016 in large part to Demand shocks. The oil price drop at the beginning of the pandemic and subsequent recovery is attributed by both models chiefly to Oil Demand. This is in line with Baumeister (2023), who finds the bulk of oil price increase starting late 2020 was driven by a strong Demand rebound, in all its components. The HS model attributes a larger contribution to Speculative Demand, particularly since 2015. For the last variable in the equations, the proxy for world Crude Oil Inventories, the largest contributions seem to be from Shock 4, the Speculative Demand Shock. This is true for both models. In this case, it is the HS

model which estimates a larger contribution of Flow Demand. In particular, during the 2008-2014 period the IG model attributes mostly to Economic Activity shocks the cumulative contributions which the HS model attributes to both these shocks and those in Oil Demand. Right after the start of the pandemic and recovery right after, the HS model also attributes to Oil Consumption Demand shocks a sizable contribution to the decrease in inventories, which the IG model estimates much smaller. It makes some sense for the Inventories of crude oil to be influenced by Consumption Demand for oil (Baumeister and Kilian, 2015), so it is plausible that recovered Consumption Demand in the post-pandemic period would contribute to the decrease in inventories.

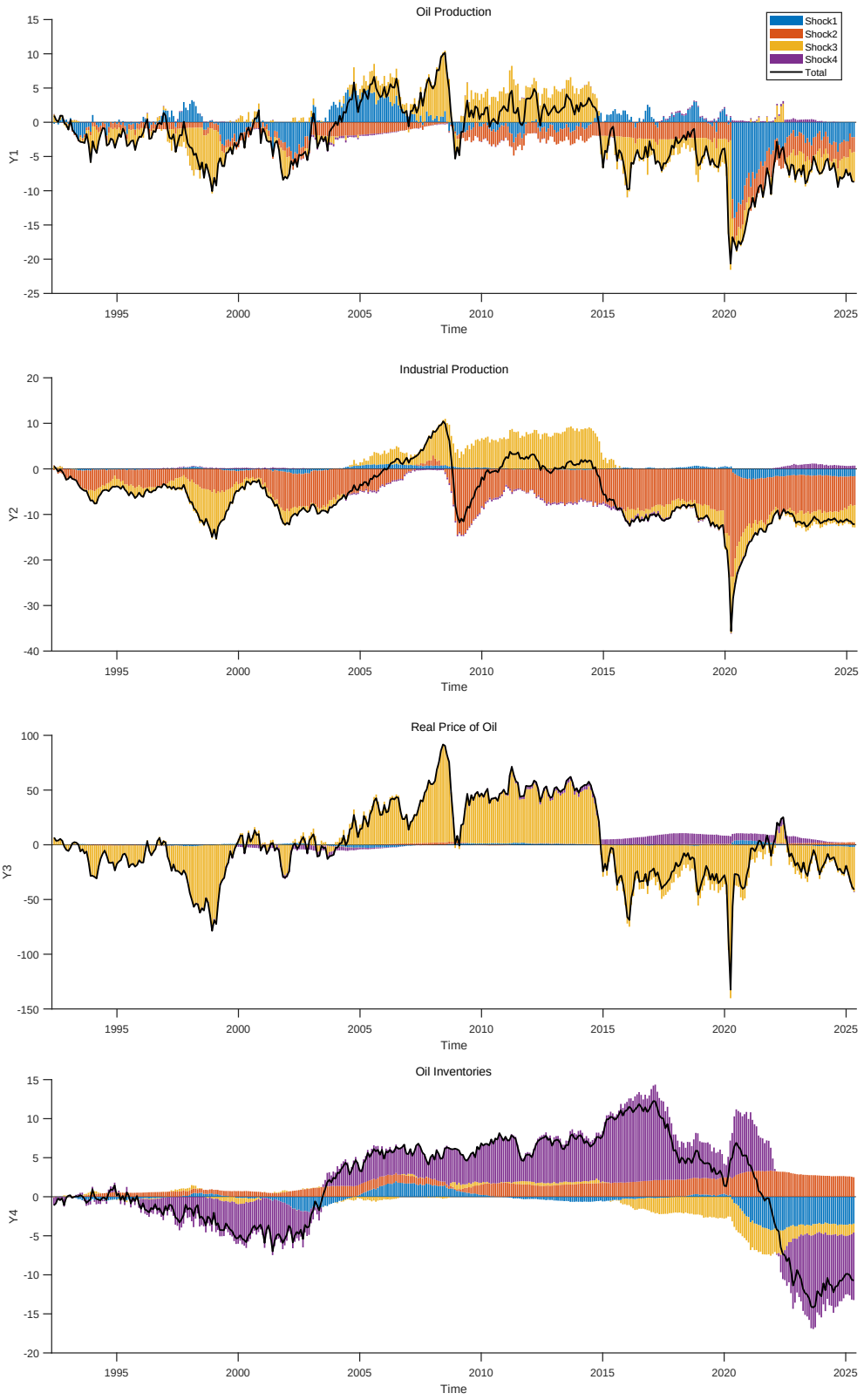


Figure 14: Historical Decompositions for all variables, computed with the HS model. Each estimated accumulated effect is computed as the median effect across all draws.

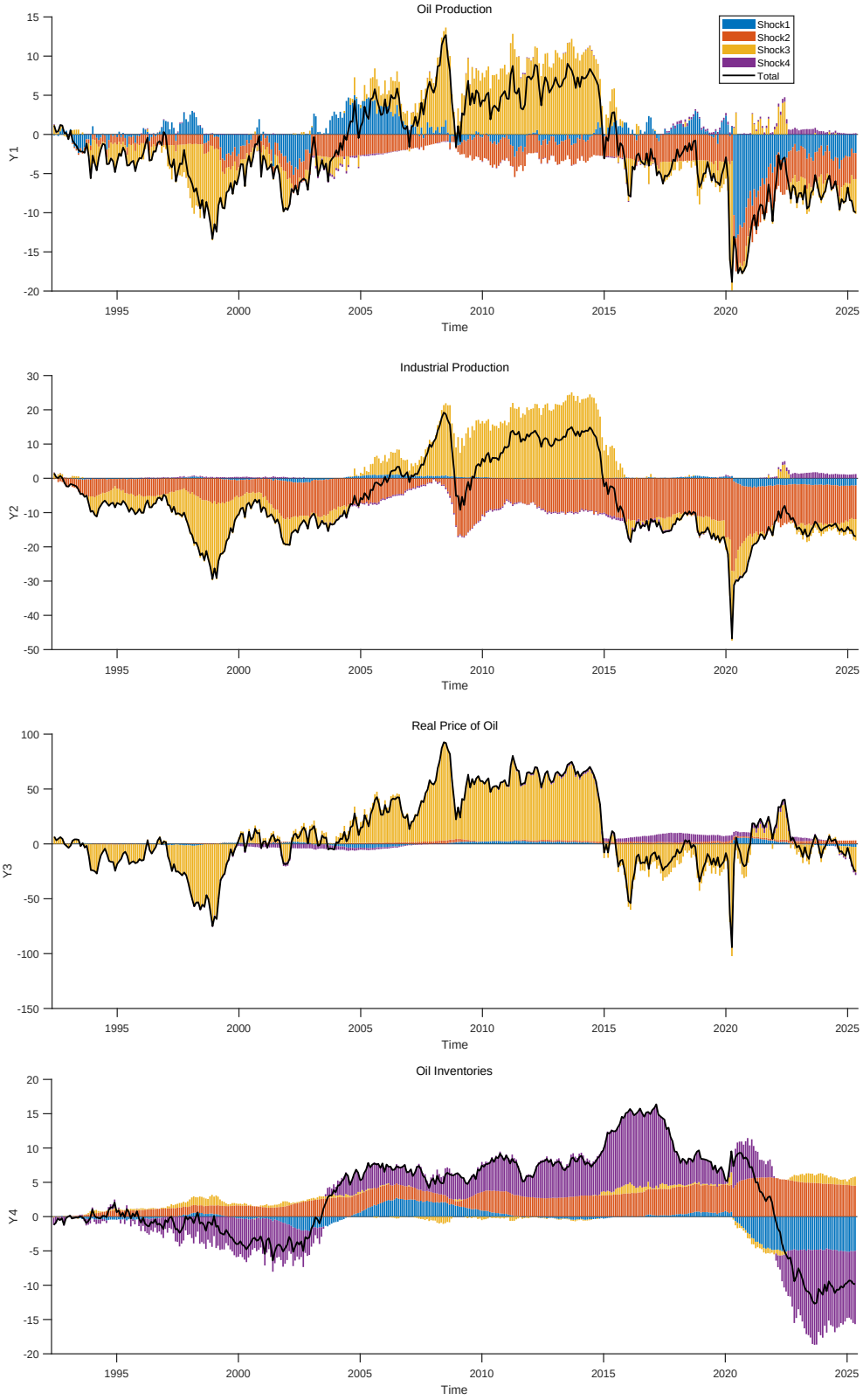


Figure 15: Historical Decompositions for all variables, computed with the IG model. Each estimated accumulated effect is computed as the median effect across all draws.

6.5 The role of the prior: Discussion

The choice to model ω_{it}^2 as time-varying, through its prior, does yield slightly different results than the standard, time constant, tight IG prior version.

The obtained IRFs are similar for both models, but the HS model estimates slightly more conservative responses all around, which can indicate it is less sensitive to extreme observations such as the pandemic.

The FEVDs estimated with the IG model seem to overestimate the importance of oil Consumption Demand to the variance of the model variables. Particularly, the finding that Economic Activity shocks have little importance to the variance of Industrial Production seems incorrect.

One can consider the relatively higher contributions of Speculative Demand to Oil Price estimated by the HS model as a sign that this model captures the underlying dynamics better, if following the rationale of Kilian and Murphy (2014), that positive Speculative Demand shocks reflect expected future shortfall of supply relative to demand for oil, so reflect an expectation of future oil price increases. In this sense, we would expect to see a contribution of Inventory Demand in the recovery period of Oil Prices, driven by an expectation that prices would increase after the low point in early 2020.

Overall, the argument can be made that the HS model, and hence the assumption of time-varying ω_{it}^2, is better at capturing structural oil market dynamics for a sample which includes the pandemic period.

However, the literature is far from unanimous regarding oil market dynamics, and not much research is yet available that looks specifically into the pandemic period in terms of Supply and Demand Shocks. Hence, it is hard to make an absolute assertion as to which model captures those dynamics best.

7 Conclusion

This thesis looked into flexible prior choices in Stochastic-Volatility identified VAR models, specifically into setting a prior such that the variance of log-volatility processes is

time-varying. The goal was to find out whether such a choice improves the VAR performance, relative to the standard time constant variance of volatility, when the variables present a large, sudden change in behavior.

A major challenge of the chosen VAR framework when applied to processes with sharp changes in volatility is that it often requires a somewhat tight prior variance for the impact matrix. This causes an incorrect scaling of the impact matrix entries, which requires normalizing the matrix. This has implications for the way the variance of the reduced-form shocks and VAR tools, like forecast error decompositions, are computed.

Simulation Studies reveal that the more flexible prior does capture these changes better in most cases, particularly in regard to the variance of model shocks. The estimates of VAR parameters and impact matrix are similar between prior choices.

In a small empirical study, both prior versions are applied to the global oil market, identifying Supply and Demand structural shocks in a 4-variable VAR(13). The flexible prior estimates sharper jumps in the log-volatility processes at the start of the pandemic, which is in line with the intuition that the pandemic start caused extreme structural shocks. However, analysis with standard VAR tools computed with each version of the model yield similar results.

The results point towards improved estimation with the flexible prior choice in a theoretical setting, but in the chosen empirical application there is not very conclusive evidence of a real improvement.

Appendices

A MCMC algorithm

This section lays out the remaining steps of posterior sampler.

The steps are reproduced here for convenience:

1. Sample $\mathbf{B}_0$
2. Sample $\boldsymbol{\alpha}$
3. Sample $\mathbf{h}$
4. Sample ω
5. Sample $\boldsymbol{\phi}$
6. Sample $\boldsymbol{\psi}$
7. Sample $\boldsymbol{z}_{\boldsymbol{\psi}}$
8. Sample $\boldsymbol{\kappa}$
9. Sample $\boldsymbol{z}_{\boldsymbol{\kappa}}$

Steps 1, 3, 4 and 5 have already been explained in the main text. The remaining are detailed below:

Step 2. The VAR coefficients are sampled row by row of $\boldsymbol{\alpha}$, $\boldsymbol{\alpha}_i, i = 1, ..., n$ as well. The same algorithm as in Chan et al. (2024), which constitutes an extension of the algorithm in Carriero et al. (2022) to the case when $\boldsymbol{B_0}$ is not (necessarily) lower triangular.

Let $\mathbf{A}_{i=0}$ be the matrix $\mathbf{A}$ from Equation 5, except for the i-th column, which is zero, i.e., $\mathbf{A}_{i=0} = (\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_{i-1}, \mathbf{0}, \boldsymbol{\alpha}_{i+1}, \ldots, \boldsymbol{\alpha}_n)$.

Rewriting Equation 1 as

$$\boldsymbol{B_0}(\mathbf{y}_t - \mathbf{A}'_{i=0}\mathbf{x}_t) = (\mathbf{B}_{0,i} \otimes \mathbf{x}'_t)\boldsymbol{\alpha}_i + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{D}_t)$$

where $\mathbf{x}_t = (1, \mathbf{y}'_{t-1}, \ldots, \mathbf{y}'_{t-p})'$ and $\mathbf{B}_{0,i}$ is the i-th column of $\boldsymbol{B_0}$. Let $\mathbf{z}^i_t = \boldsymbol{B_0}(\mathbf{y}_t - \mathbf{A}'_{i=0}\mathbf{x}_t)$ and $\mathbf{z}^i = (\mathbf{z}^i_1, \ldots, \mathbf{z}^i_T)'$. Then,

$$\mathbf{z}^i = \mathbf{W}^i\boldsymbol{\alpha}_i + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{D}),$$

with $\mathbf{W}^i = \mathbf{X} \otimes \mathbf{B}_{0,i}$ and $\mathbf{D} = diag(\mathbf{D}_1, \ldots, \mathbf{D}_T)$.

The prior on $\boldsymbol{\alpha}_i$ is conditionally Normal given $\boldsymbol{\psi}_i$, the local variance component. So, by standard linear regression results,

$$(\boldsymbol{\alpha}_i \mid \mathbf{y}, \boldsymbol{B_0}, \boldsymbol{\alpha}_{-i}, \mathbf{h}, \boldsymbol{\psi}_i, \boldsymbol{\kappa}) \sim \mathcal{N}(\hat{\boldsymbol{\alpha}}_i, \mathbf{K}^{-1}_{\boldsymbol{\alpha}_i}),$$

where $\boldsymbol{\alpha}_{-i} = (\boldsymbol{\alpha}'_1, \ldots, \boldsymbol{\alpha}'_{i-1}, \boldsymbol{\alpha}'_{i+1}, \ldots, \boldsymbol{\alpha}'_n)'$ and $\mathbf{K}_{\boldsymbol{\alpha}_i} = \mathbf{V}^{-1}_{\boldsymbol{\alpha}_i} + \mathbf{W}^{i'}\mathbf{D}^{-1}\mathbf{W}^i$, $\hat{\boldsymbol{\alpha}}_i = \mathbf{K}^{-1}_{\boldsymbol{\alpha}_i}\left(\mathbf{V}^{-1}_{\boldsymbol{\alpha}_i}\mathbf{m}_i + \mathbf{W}^{i'}\mathbf{D}^{-1}\mathbf{z}\right.$ with $\mathbf{V}_{\boldsymbol{\alpha}_i} = diag(C_{i,1}, \kappa_{i,2}\psi_{i,2}C_{i,2}, \ldots, \kappa_{i,k}\psi_{i,k}C_{i,k})$ and $\mathbf{m}_i = (m_{i,1}, \ldots, m_{i,k})'$.

Step 6. This is simplified by the fact that $\psi_{i,j}$ are conditionally independent, meaning they can be sampled separately. Their respective distributions are

$$p(\psi_{i,j}|\alpha_{i,j}, \kappa, z_{\psi_{i,j}}) \propto \psi^{-2}_{i,j} \exp\left(-\frac{1}{\psi_{i,j}}\left(z^{-1}_{\psi_{i,j}} + \frac{\alpha^2_{i,j}}{2\kappa_{i,j}C_{i,j}}\right)\right)$$

which corresponds to the kernel of an inverse-Gamma distribution, meaning that

$$(\psi_{i,j}|\alpha_{i,j}, \kappa, z_{\psi_{i,j}}) \sim \mathcal{IG}\left(1, z^{-1}_{\psi_{i,j}} + \frac{\alpha^2_{i,j}}{2\kappa_{i,j}C_{i,j}}\right)$$

Step 7. This is a straightforward step. By definition, $z_{\psi_{ij}} \sim \mathcal{IG}(1, 1 + \psi^{-1}_{ij})$, so it is simply sampled from this distribution directly.

Step 8. As mentioned previously, $\kappa_{i,j} = \kappa_1$ for "own lags" and κ_2 for "other lags". For ease of notation, define groups of indices (i, j) of the autoregressive coefficients $\alpha_{i,j}$, S_{κ_1} and S_{κ_2} such that $S_{\kappa_1} = \{(i, j) : \alpha_{i,j}$ is coefficient on own lag $\}$ and $S_{\kappa_2} = \{(i, j) : \alpha_{i,j}$ is coefficient on other lag $\}$. Note that S_{κ_1} has np elements and S_{κ_2} has $(n-1)np$. Then,

the conditional posterior of κ_1 can be written as

$$p(\kappa_1|\boldsymbol{\alpha}, \boldsymbol{\psi}, \boldsymbol{z_{\kappa_1}}) \propto \prod_{(i,j)\in S_{\kappa_1}} \kappa_1^{-\frac{1}{2}} \exp\left\{-\frac{1}{2\kappa_1 C_{i,j}\psi_{i,j}}\alpha_{i,j}^2\right\} \kappa_1^{-\frac{3}{2}} e^{-\frac{1}{\kappa_1 z_{\kappa_1}}}$$
$$= \kappa_1^{-\left(\frac{np+1}{2}+1\right)} \exp\left\{\frac{1}{\kappa_1}\left(z_{\kappa_1}^{-1} + \sum_{(i,j)\in S_{\kappa_1}} \frac{\alpha_{i,j}^2}{2\psi_{i,j}C_{i,j}}\right)\right\}$$

which is the kernel of the inverse Gamma distribution, meaning we can sample κ_1 from $\mathcal{IG}\left(\frac{np+1}{2}, z_{\kappa_1}^{-1} + \sum_{(i,j)\in S_{\kappa_1}} \frac{\alpha_{i,j}^2}{2\psi_{i,j}C_{i,j}}\right)$.

Similarly,

$$(\kappa_2|\alpha, \psi, z_{\kappa_2}) \sim \mathcal{IG}\left(\frac{(n-1)np+1}{2}, z_{\kappa_2}^{-1} + \sum_{(i,j)\in S_{\kappa_2}} \frac{\alpha_{i,j}^2}{2\psi_{i,j}C_{i,j}}\right)$$

Step 9. Similarly to Step 7, sample directly $z_{\kappa_\ell} \sim \mathcal{IG}(1, 1 + \kappa_\ell^{-1}), \ell = 1, 2$.

B Scaling Issue: Alternative impact matrix

Here are presented the densities for a similar simulation exercise as the one described in Section 4, but with a different $\boldsymbol{B_0}$, with different ratios to the diagonal:

$$\mathbf{B_0} = \begin{bmatrix} 3 & 1.5 & -0.2 \\ 0.5 & 2 & 0.4 \\ 0.8 & -0.6 & 2.5 \end{bmatrix}$$

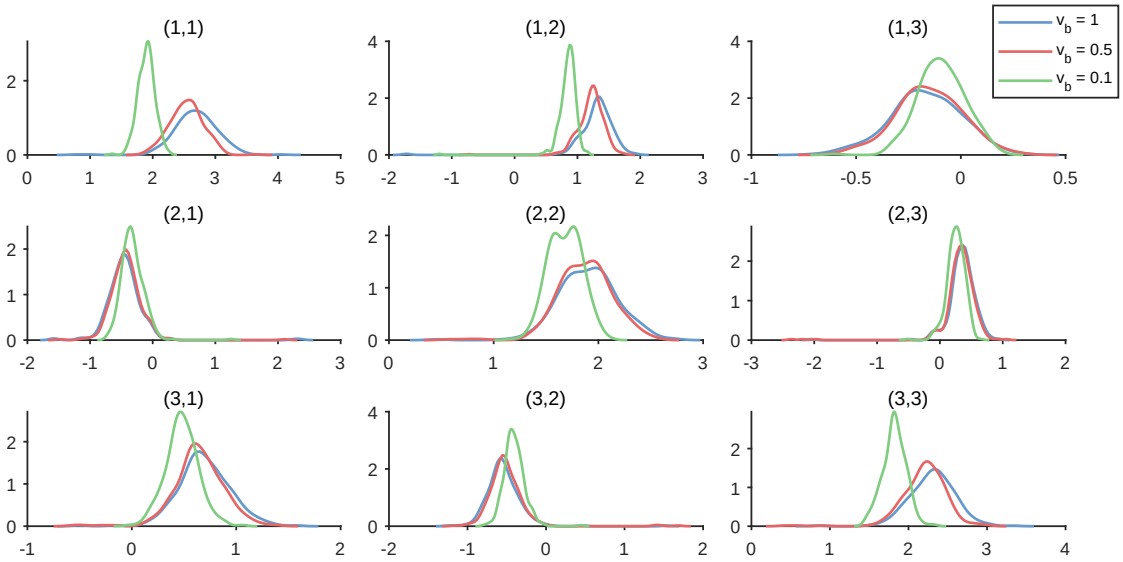


Figure B.1: Densities of estimated B0 entries with different v_b across $R = 300$ datasets, for the alternative $\boldsymbol{B_0}$. The estimate is the median of draws. The plots are organized like the entries they correspond to, so the first plot is the $(1, 1)$ entry of $\boldsymbol{B_0}$, etc.

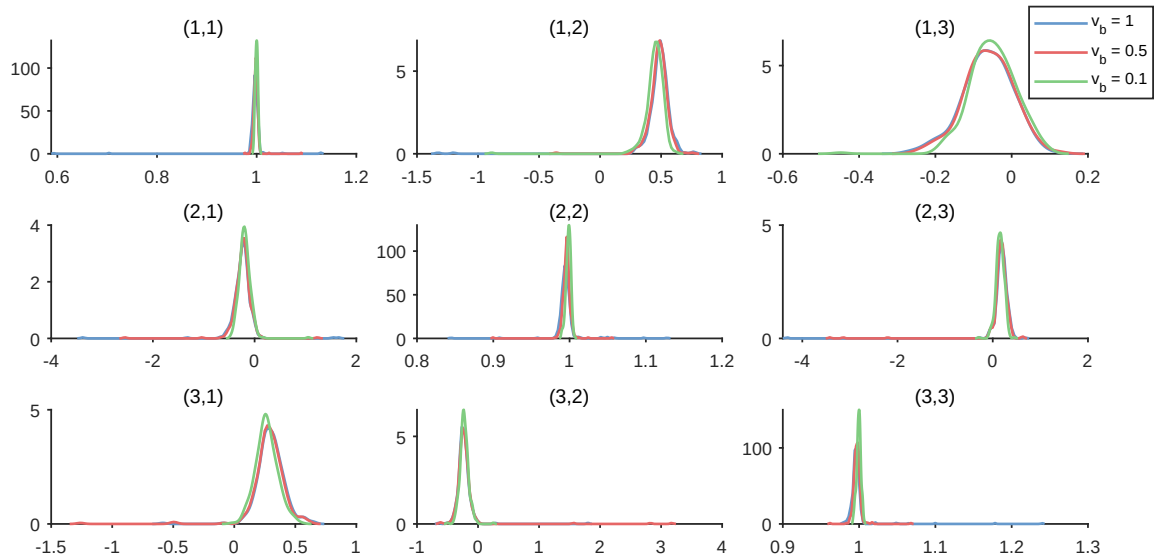


Figure B.2: Densities of estimated B0 entries with different v_b across $R = 300$ datasets, for the alternative $\boldsymbol{B_0}$. The estimate is the median of draws. The plots are organized like the entries they correspond to.

C Figures

Scaling Issue

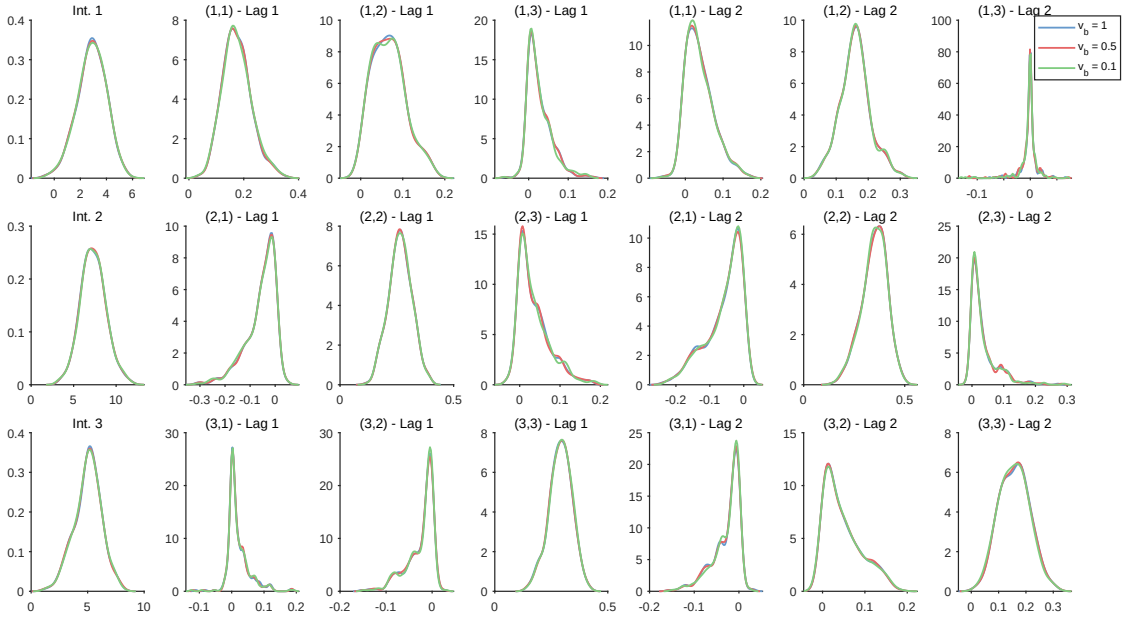
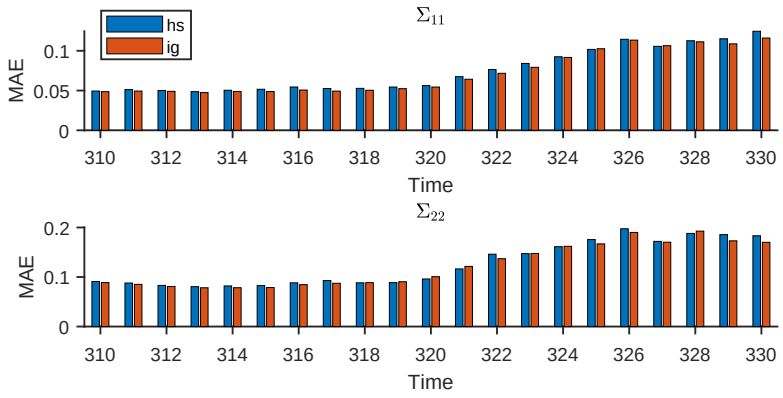
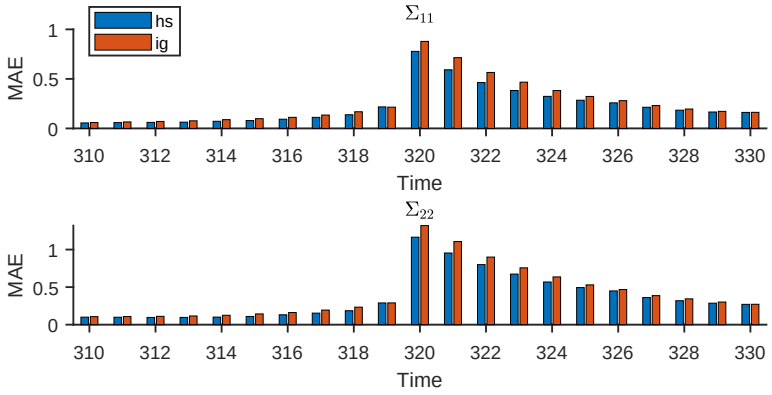


Figure C.1: Densities of estimated $\boldsymbol{\alpha}$ entries with different v_b across $R = 300$ datasets. The estimate is the median of draws. The first column presents the intercepts, the next 3 columns the entries of $\boldsymbol{A}_1$, organized as the respective entries, and the last 3 columns present the entries of $\boldsymbol{A}_2$

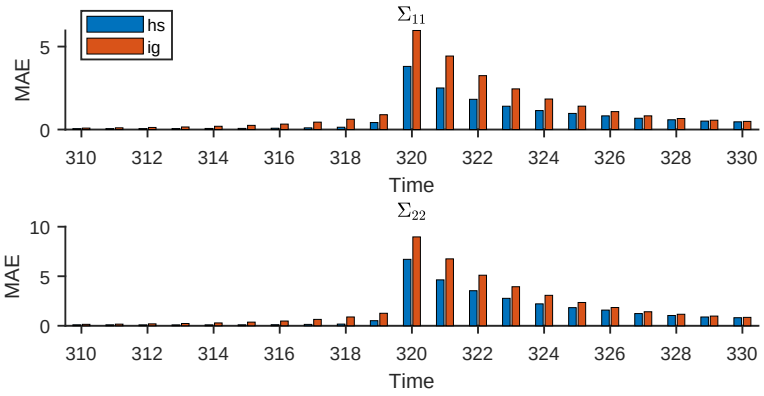
Simulation Study



(a) DGP1

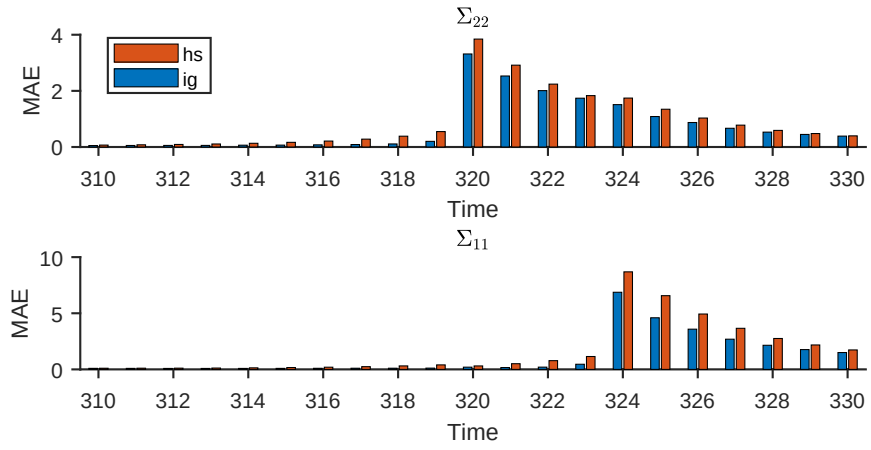


(b) DGP2

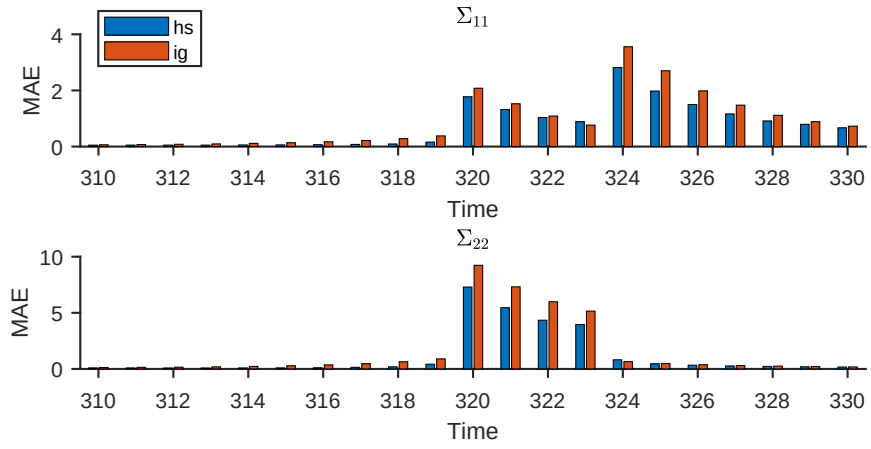


(c) DGP3

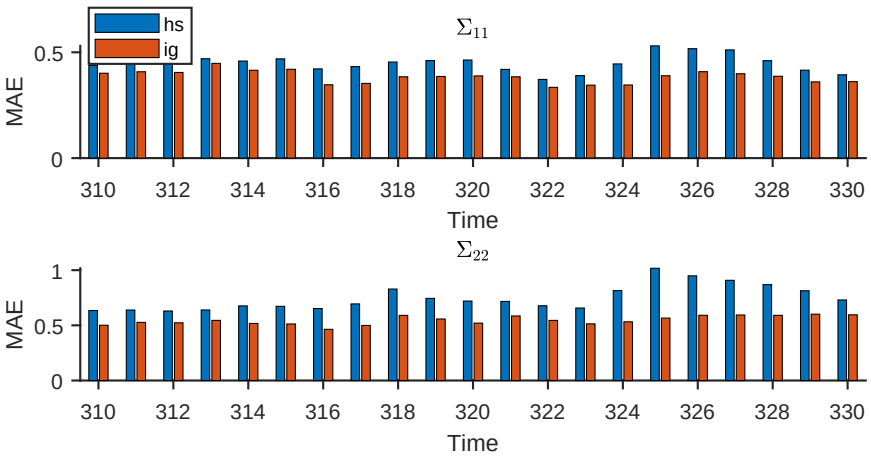
Figure C.2: Individual MAE values for the $\boldsymbol{\Sigma}_t = \boldsymbol{B}_{\mathbf{0}}^{-1}\boldsymbol{D}_t(\boldsymbol{B}_{\mathbf{0}}^{-1})'$ diagonal entries, for 10 periods before and after $T = 320$, for DGP1-3. The estimates taken for the MAE computation are the medians of draws the respective entry of $\boldsymbol{\Sigma}_t$.



(a) DGP4

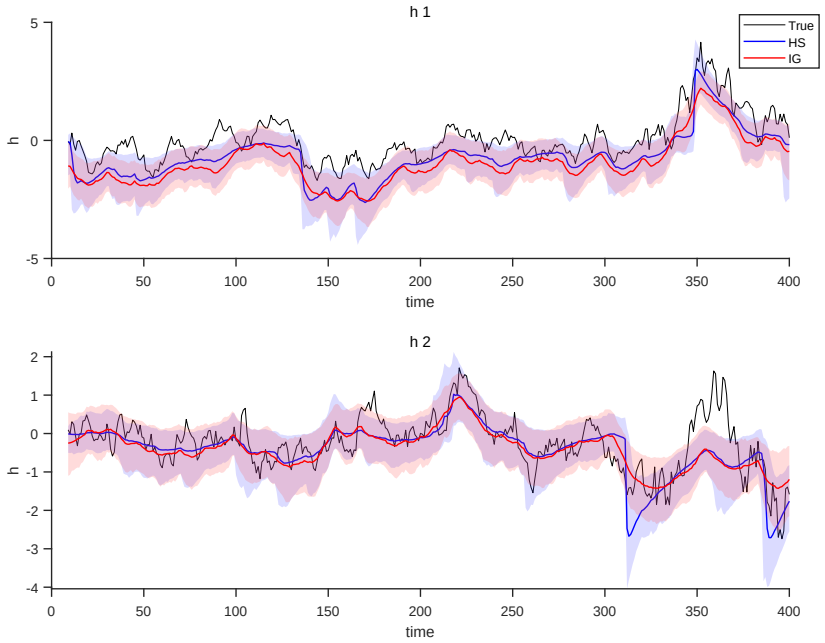


(b) DGP5

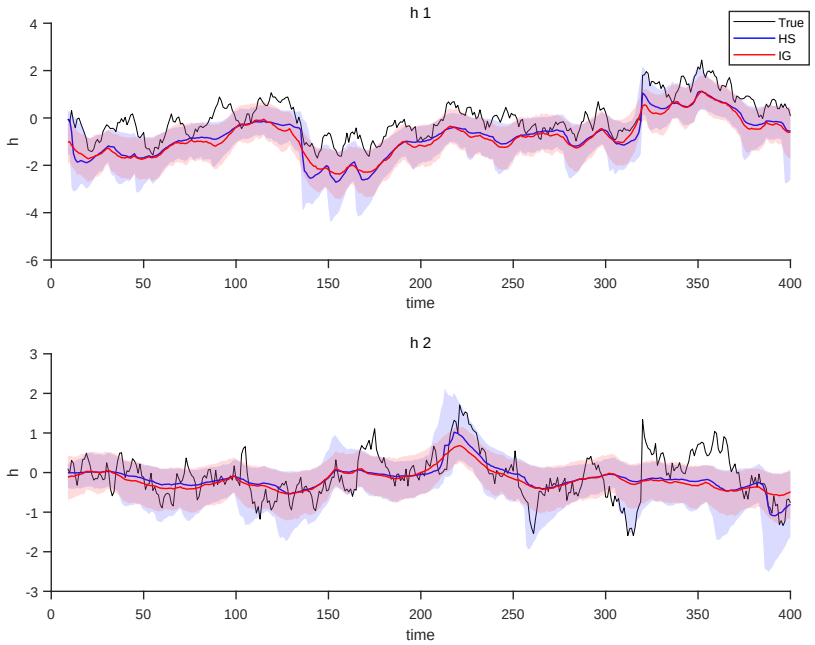


(c) DGP6

Figure C.3: Individual MAE values for the $\boldsymbol{\Sigma}_t = \boldsymbol{B}_0^{-1}\boldsymbol{D}_t(\boldsymbol{B}_0^{-1})'$ diagonal entries, for 10 periods before and after $T = 320$, for DGP4-6. The estimates taken for the MAE computation are the medians of draws the respective entry of $\boldsymbol{\Sigma}_t$.

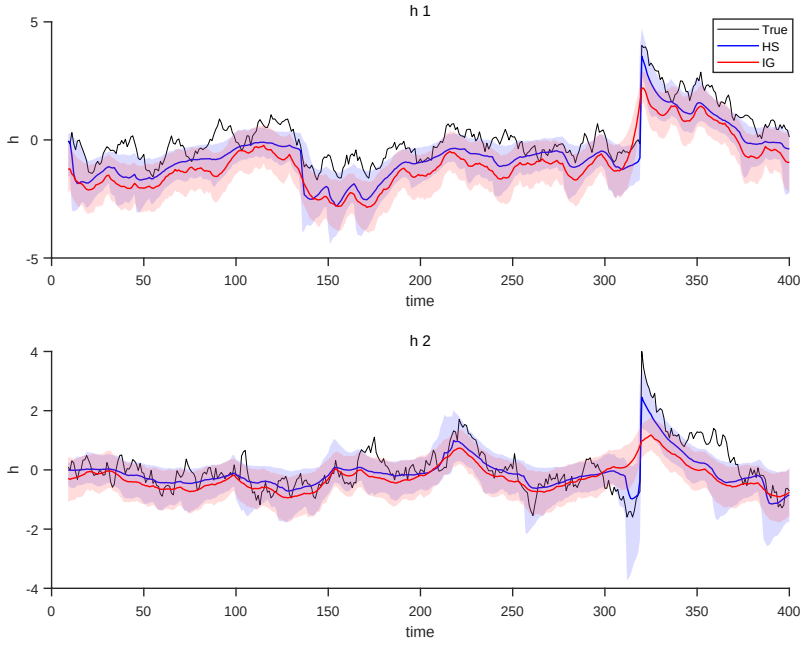


(a) DGP1

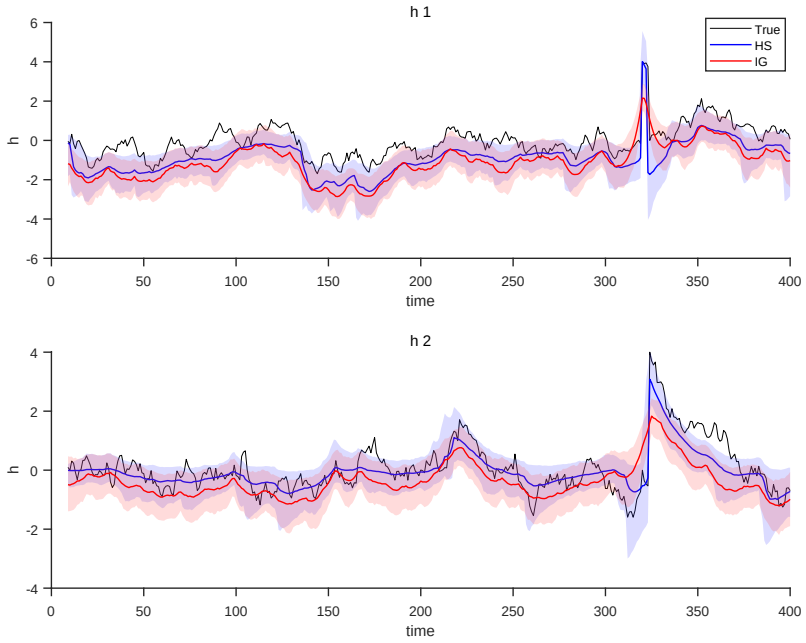


(b) DGP2

Figure C.4: 250th generated log-volatility process from DGP1-2 and respective estimates with HS and IG model. The full lines represent the median of draws, and the shaded areas the 90% credible regions for each estimate.

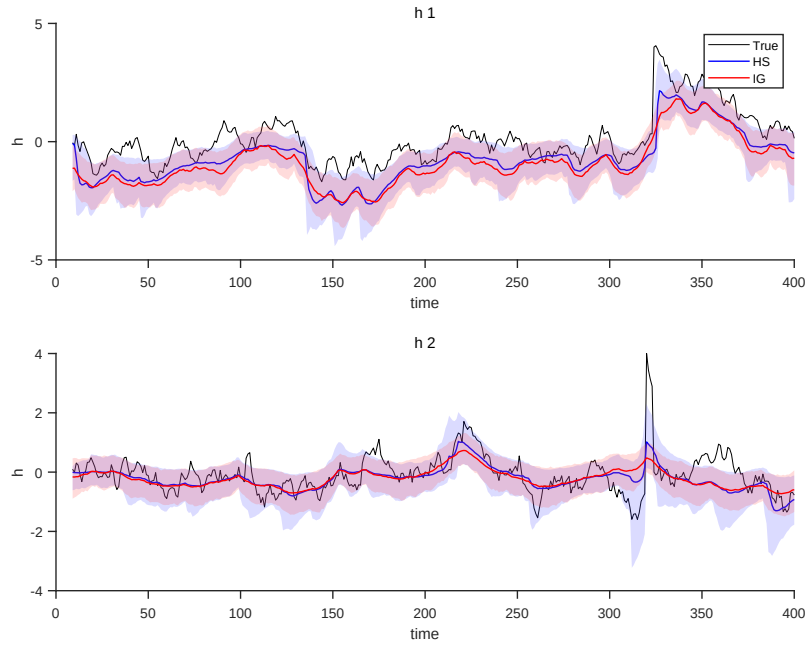


(a) DGP3

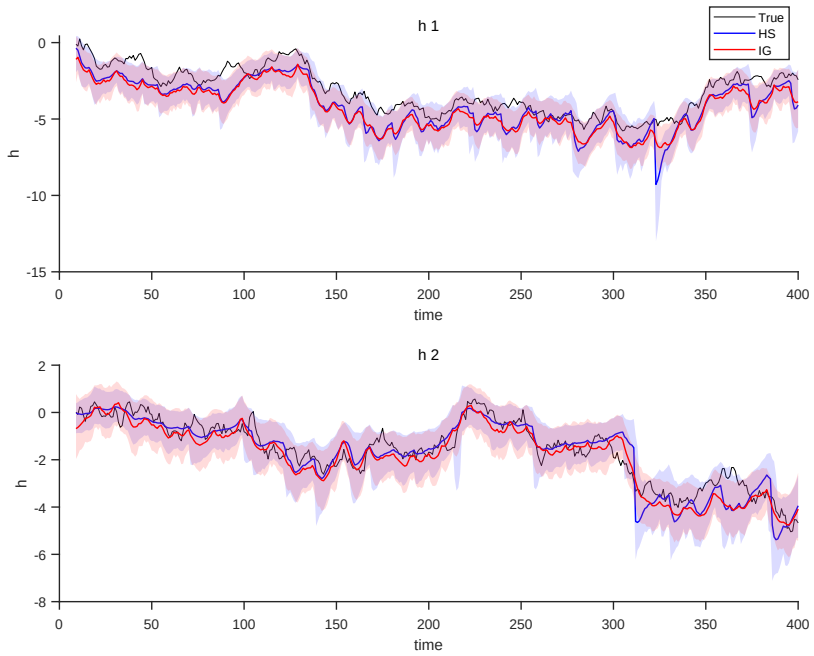


(b) DGP4

Figure C.5: 250th generated log-volatility process from DGP3-4 and respective estimates with HS and IG model. The full lines represent the median of draws, and the shaded areas the 90% credible regions for each estimate.



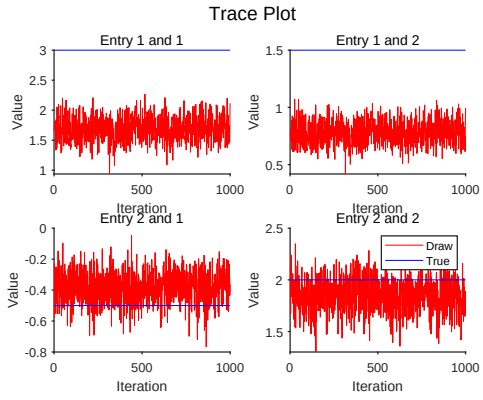
(a) DGP5



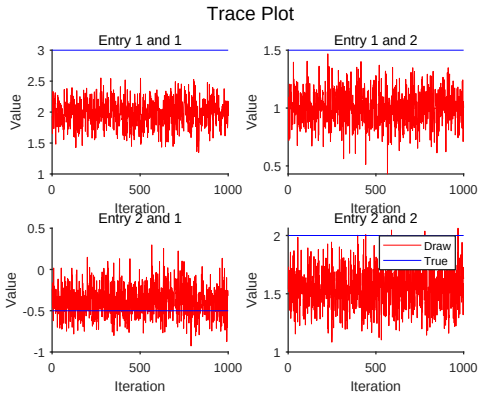
(b) DGP6

Figure C.6: 250th generated log-volatility process from DGP5-6 and respective estimates with HS and IG model. The full lines represent the median of draws, and the shaded areas the 90% credible regions for each estimate.

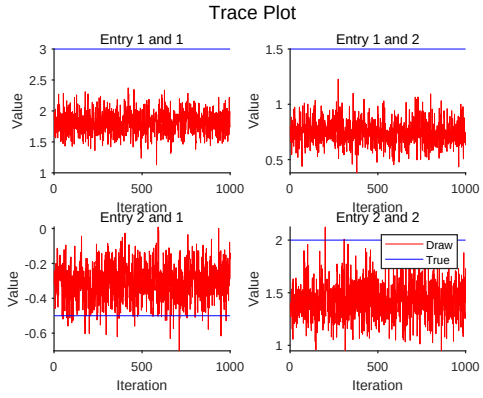
Convergence



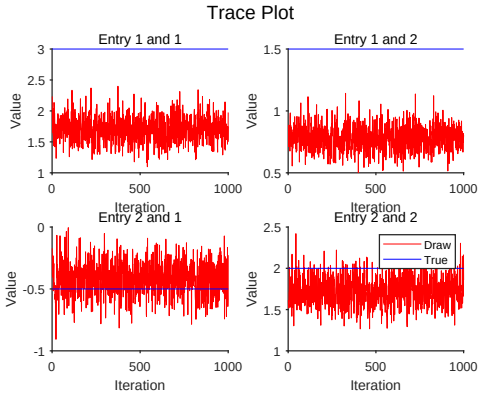
(a) DGP1



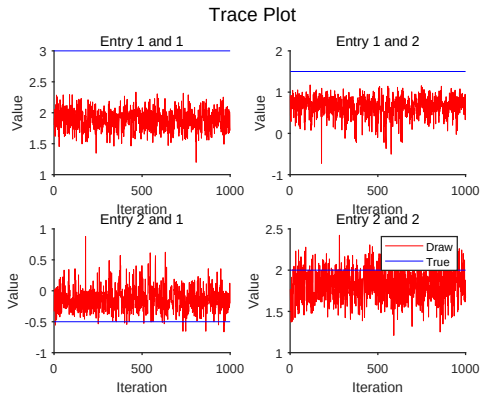
(b) DGP2



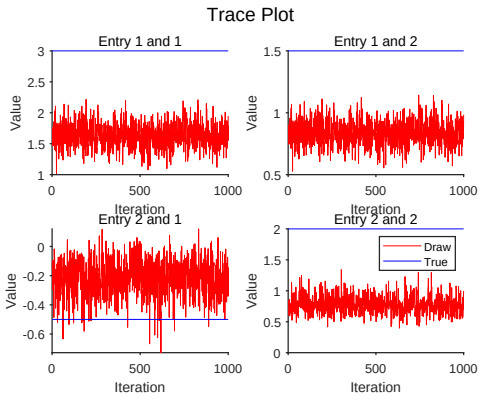
(c) DGP3



(d) DGP4

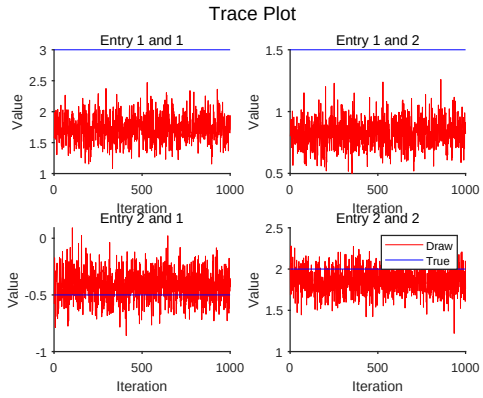


(e) DGP5

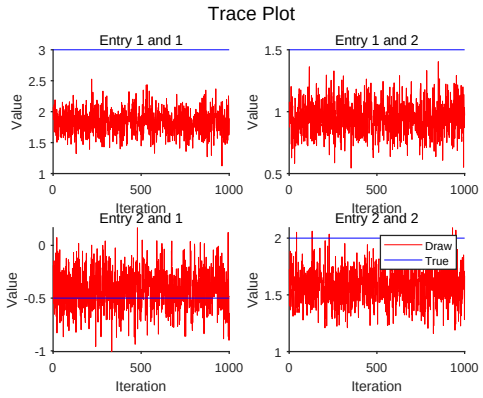


(f) DGP6

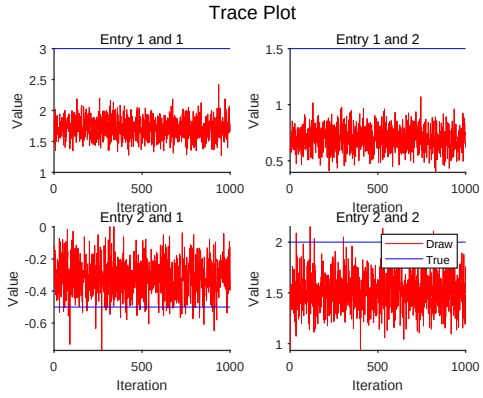
Figure C.7: Trace plots of draws of $\boldsymbol{B_0}$ entries for all DGPs, with the HS model. The chains are represented in red, and the true entries of $\boldsymbol{B_0}$ are marked in blue. The draws are not normalized. In each figure, the plots are organized like the entries of the matrix they correspond to.



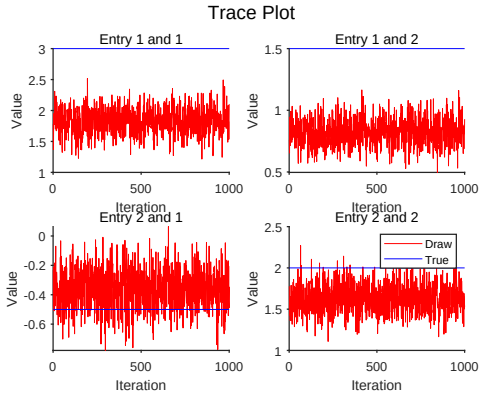
(a) DGP1



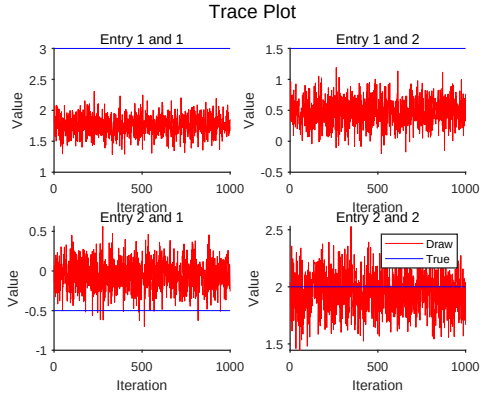
(b) DGP2



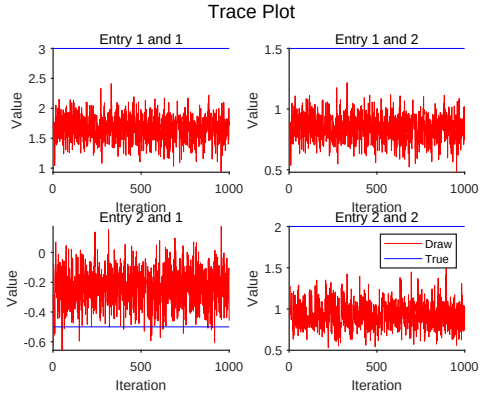
(c) DGP3



(d) DGP4



(e) DGP5



(f) DGP6

Figure C.8: Trace plots of draws of $\boldsymbol{B_0}$ entries for all DGPs, with the IG model. The chains are represented in red, and the true entries of $\boldsymbol{B_0}$ are marked in blue. The draws are not normalized. In each figure, the plots are organized like the entries of the matrix they correspond to.

Empirical Analysis

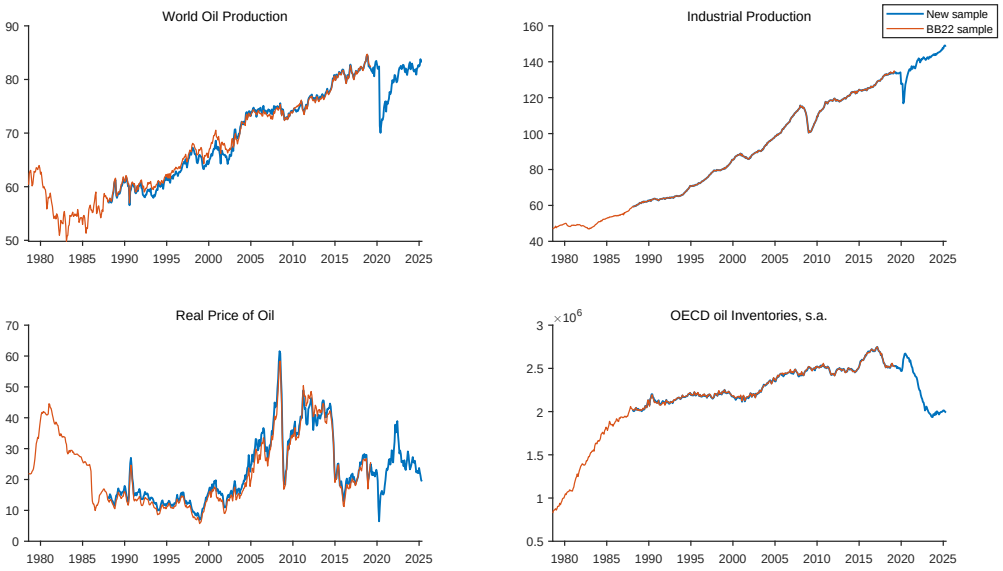


Figure C.9: Oil market variables, both the constructed sample (New) and the one from Braun and Brüggemann (2022) (BB22), for the periods used in the estimation.

D Tables

Simulation Study

HS

	$\boldsymbol{\alpha}$	$\boldsymbol{B_0}$	$\boldsymbol{\sigma}_{11}$	$\boldsymbol{\sigma}_{12} = \boldsymbol{\sigma}_{21}$	$\boldsymbol{\sigma}_{22}$
DGP1	0.196	0.005	0.045	0.032	0.117
DGP2	0.212	0.009	0.015	0.012	0.050
DGP3	0.209	0.010	0.122	0.102	0.470
DGP4	0.190	0.004	0.120	0.135	0.499
DGP5	0.171	0.004	0.101	0.120	0.440
DGP6	0.199	0.006	1.100	1.353	5.111

IG

	$\boldsymbol{\alpha}$	$\boldsymbol{B_0}$	$\boldsymbol{\sigma}_{11}$	$\boldsymbol{\sigma}_{12} = \boldsymbol{\sigma}_{21}$	$\boldsymbol{\sigma}_{22}$
DGP1	0.196	0.005	0.034	0.032	0.118
DGP2	0.212	0.010	0.016	0.013	0.052
DGP3	0.210	0.011	0.242	0.109	0.620
DGP4	0.194	0.005	0.123	0.168	0.578
DGP5	0.178	0.005	0.124	0.169	0.585
DGP6	0.194	0.005	0.757	0.743	2.813

Table D.1: Average MSE values for $\boldsymbol{B_0}$, $\boldsymbol{\alpha}$, and entries of $\boldsymbol{\Sigma}_t = \boldsymbol{B_0}^{-1}\boldsymbol{D}_t(\boldsymbol{B_0}^{-1})'$, for each DGP. The estimate used to compute the MSE is the median of draws of each entry. The first panel contains the values computed with the HS model, the second with the IG model.

HS

Parameter	α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8	α_9	α_{10}	b_{11}	b_{12}	b_{21}	b_{22}
DGP1	0.726	0.046	0.032	0.048	0.034	0.835	0.047	0.041	0.054	0.043	0.002	0.038	0.069	0.003
DGP2	0.775	0.048	0.033	0.051	0.035	0.869	0.051	0.041	0.059	0.043	0.002	0.053	0.094	0.003
DGP3	0.779	0.048	0.033	0.051	0.036	0.847	0.051	0.041	0.059	0.043	0.002	0.054	0.098	0.003
DGP4	0.700	0.045	0.031	0.048	0.032	0.849	0.048	0.041	0.057	0.043	0.002	0.031	0.065	0.002
DGP5	0.737	0.045	0.032	0.048	0.034	0.739	0.043	0.039	0.048	0.040	0.002	0.037	0.060	0.003
DGP6	0.667	0.041	0.034	0.044	0.035	0.813	0.043	0.043	0.049	0.044	0.004	0.036	0.051	0.006

IG

Parameter	α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8	α_9	α_{10}	b_{11}	b_{12}	b_{21}	b_{22}
DGP1	0.721	0.045	0.032	0.048	0.033	0.847	0.047	0.041	0.054	0.042	0.002	0.039	0.071	0.002
DGP2	0.765	0.047	0.033	0.050	0.035	0.879	0.050	0.041	0.059	0.043	0.003	0.054	0.096	0.002
DGP3	0.772	0.048	0.033	0.050	0.035	0.857	0.050	0.041	0.058	0.043	0.002	0.053	0.101	0.003
DGP4	0.713	0.045	0.031	0.049	0.032	0.856	0.050	0.040	0.059	0.044	0.002	0.034	0.073	0.003
DGP5	0.752	0.045	0.032	0.049	0.034	0.765	0.043	0.041	0.048	0.041	0.002	0.040	0.066	0.003
DGP6	0.655	0.039	0.034	0.044	0.035	0.817	0.042	0.043	0.049	0.044	0.003	0.035	0.048	0.005

Table D.2: Individual MAE values for entries of $\boldsymbol{B_0}$ and $\boldsymbol{\alpha}$, for each DGP. The b_{ij} columns refer to the ij-th normalized entry of the $\boldsymbol{B_0}$ matrix, and the α_i to the i-th entry of $\boldsymbol{\alpha}$. The estimate used to compute the MAE is the median of draws of each entry. The first panel contains the values computed with the HS model, the second with the IG model.

HS

Parameter	α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8	α_9	α_{10}	b_{11}	b_{12}	b_{21}	b_{22}
DGP1	0.816	0.003	0.001	0.003	0.002	1.115	0.003	0.003	0.004	0.003	0.000	0.003	0.008	0.000
DGP2	0.906	0.004	0.002	0.004	0.002	1.191	0.004	0.003	0.005	0.003	0.000	0.005	0.014	0.000
DGP3	0.917	0.004	0.002	0.004	0.002	1.144	0.004	0.003	0.005	0.003	0.000	0.005	0.015	0.000
DGP4	0.742	0.003	0.001	0.003	0.002	1.137	0.003	0.003	0.005	0.003	0.000	0.002	0.007	0.000
DGP5	0.828	0.003	0.001	0.003	0.002	0.858	0.003	0.003	0.004	0.003	0.000	0.002	0.006	0.000
DGP6	0.789	0.003	0.002	0.003	0.002	1.172	0.004	0.003	0.005	0.003	0.000	0.005	0.007	0.000

IG

Parameter	α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8	α_9	α_{10}	b_{11}	b_{12}	b_{21}	b_{22}
DGP1	0.805	0.003	0.001	0.003	0.002	1.127	0.003	0.003	0.005	0.003	0.000	0.003	0.008	0.000
DGP2	0.887	0.003	0.002	0.004	0.002	1.206	0.004	0.003	0.005	0.003	0.000	0.005	0.015	0.000
DGP3	0.899	0.004	0.002	0.004	0.002	1.173	0.004	0.003	0.005	0.003	0.000	0.005	0.017	0.000
DGP4	0.761	0.003	0.001	0.003	0.002	1.150	0.004	0.003	0.005	0.003	0.000	0.002	0.009	0.000
DGP5	0.854	0.003	0.002	0.003	0.002	0.908	0.003	0.003	0.004	0.003	0.000	0.003	0.007	0.000
DGP6	0.751	0.002	0.002	0.003	0.002	1.169	0.003	0.003	0.005	0.003	0.000	0.004	0.007	0.000

Table D.3: Individual MSE values for entries of $\boldsymbol{B_0}$ and $\boldsymbol{\alpha}$, for each DGP. The b_{ij} columns refer to the ij-th normalized entry of the $\boldsymbol{B_0}$ matrix, and the α_i to the i-th entry of $\boldsymbol{\alpha}$. The estimate used to compute the MSE is the median of draws of each entry. The first panel contains the values computed with the HS model, the second with the IG model.

Empirical Analysis

HR20				
Shock	1	2	3	4
Supply	0.149	-0.160	-0.095	-0.001
Flow Demand	-0.188	0.107	0.081	0.004
Inventory Demand	-0.210	0.048	0.129	0.007
Residual	0.103	-0.174	-0.037	-0.005
BH19				
Shock	1	2	3	4
Supply	0.122	-0.174	-0.072	-0.001
Flow Demand	-0.210	0.077	0.104	0.005
Inventory Demand	-0.209	0.045	0.143	0.008
Residual	0.141	-0.154	-0.055	-0.006

Table D.4: Correlation coefficients between the shock series from Braun and Brüggemann (2022), without the IV restriction and with the HR20 (first panel) and BH19 (second panel) elasticity restrictions, and the estimated structural shocks with the IG model, new sample.

HS				
Shock	1	2	3	4
Oil Supply	0.623	-0.015	-0.578	0.168
Consump. Demand	0.179	0.023	0.829	-0.278
Invent. Demand	0.052	-0.086	0.067	0.747
Econ. Activity	-0.019	0.850	0.214	0.098
IG				
Shocks	1	2	3	4
Oil Supply	0.627	-0.005	-0.577	0.151
Consump. Demand	0.168	0.013	0.828	-0.279
Invent. Demand	0.081	-0.096	0.067	0.742
Econ. Activity	-0.010	0.852	0.216	0.108

Table D.5: Correlation coefficients between the shock series from Baumeister and Hamilton (2019) and the estimated structural shocks, with the HS model (first panel) and the IG model (second panel), old sample.

HR20

Shock	1	2	3	4
Supply	0.027	0.120	0.007	0.014
Flow Demand	-0.031	-0.120	-0.048	-0.028
Inventory Demand	-0.057	-0.011	0.044	-0.022
Residual	-0.056	0.095	0.111	-0.064

BH19

Shock	1	2	3	4
Supply	0.009	0.148	0.036	0.004
Flow Demand	-0.041	-0.095	-0.020	-0.032
Inventory Demand	-0.059	-0.003	0.066	-0.020
Residual	-0.003	0.130	0.097	-0.021

Table D.6: Correlation coefficients between the shock series from Braun and Brüggemann (2022), without the IV restriction and with the HR20 (first panel) and BH19 (second panel) elasticity restrictions, and the estimated structural shocks with the HS model, old sample.

HR20

Shock	1	2	3	4
Supply	0.023	0.112	-0.002	0.013
Flow Demand	-0.027	-0.111	-0.041	-0.028
Inventory Demand	-0.051	-0.008	0.040	-0.020
Residual	-0.056	0.088	0.096	-0.061

BH19

Shock	1	2	3	4
Supply	0.006	0.138	0.023	0.004
Flow Demand	-0.037	-0.088	-0.016	-0.031
Inventory Demand	-0.053	-0.000	0.062	-0.018
Residual	-0.006	0.120	0.084	-0.020

Table D.7: Correlation coefficients between the shock series from Braun and Brüggemann (2022), without the IV restriction and with the HR20 (first panel) and BH19 (second panel) elasticity restrictions, and the estimated structural shocks with the IG model, old sample.

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